



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

FLOW INSTRUMENT PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0031
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 50

TITLE :

FLOW INSTRUMENT PROCESS DATA SHEET

(CDN UNIT)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-0031
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

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F1	JUN 25, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
A1	APR 18, 2014	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	- S.W.Lee
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REV. NO.	DATE	DESCRIPTION		PREP'N	REVIEW	QA REVIEW APPROVAL
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Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
		MAY 10, 2016	MAY 10, 2016	-		MAY 10, 2016

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		PROCESS SPECIFICATION			
PTT PHENOL TRAIN II PROJECT JOB NO. 120117		Doc. No. : 14780-8120-PS-0031		Rev. Z1	Sheet 2 / 50
Record of Revision					
NO.	REV.	PAGE NO.	DESCRIPTION		
1	O1	GENERAL	THE INDEX OF FLOW INSTRUMENTS HAS BEEN ADDED.		
2	O1	GENERAL	DESIGN TEMPERATURE AND DESIGN PRESSURE HAVE BEEN ADDED TO DEVELOP DETAILED ENGINEERING.		
3	O1	GENERAL	PROCESS DATA SHEETS OF THE ADDITIONAL FLOW INSTRUMENTS (NOT SPECIFIED IN UOP DATA SHEET)		
			HAVE BEEN ADDED.		
4	O1	GENERAL	PROCESS DATA SHEETS OF RESTRICTION ORIFICES HAVE BEEN ADDED.		
5	O1	GENERAL	TAG NAMES OF THE ASSOCIATED FLOW INSTRUMENTS HAVE BEEN REVISED IN ACCORDANCE WITH P&ID.		
6	O1	10	TAG NAME OF FT-23-1204A/B/C HAVE BEEN REVISED TO FXT-23-1204A/B/C		
			AS PER THE RESULT OF P&ID CLARIFICATION MEETING (15 JUL,13 ~ 9 AUG, 13).		
7	O1	10	TAG NAME OF FT-23-1803, 1804 HAVE BEEN REVISED TO FXT-23-1803, 1804		
			AS PER THE RESULT OF P&ID CLARIFICATION MEETING (15 JUL,13 ~ 9 AUG, 13).		
8	O1	14	TAG NAME OF FT-23-0401A/B/C HAVE BEEN REVISED TO FXT-23-0401A/B/C		
			AS PER THE RESULT OF P&ID CLARIFICATION MEETING (15 JUL,13 ~ 9 AUG, 13).		
9	O1	40	TAG NAME OF FT-23-1501, 1601, 1602 HAVE BEEN REVISED TO FXT-23-1501, 1601, 1602		
			AS PER THE RESULT OF P&ID CLARIFICATIONMEETING (15 JUL,13 ~ 9 AUG, 13).		
10	O1	43	TAG NAME OF FSL-23-1203 HAS BEEN REVISED TO FXSL-23-1203 AS PER THE RESULT OF P&ID CLARIFICATION		
			MEETING (15 JUL,13 ~ 9 AUG, 13).		
11	O1	44	TAG NAME OF FSL-23-1206 HAS BEEN REVISED TO FXSL-23-1206 AS PER THE RESULT OF P&ID CLARIFICATION		
			MEETING (15 JUL,13 ~ 9 AUG, 13).		
12	A1	GENERAL	FLOW RATE UNIT HAVE BEEN REVISED IN ACCORDANCE WITH UOP FLOW INSTRUMENT DATA SHEET		
			(VAPOR : NM3/HR, LIQUID : M3/HR, STEAM & MASS FLOW : KG/HR).		
13	A1	GENERAL	INFORMATION FOR ROs HAVE BEEN REVISED AS PER PMT'S COMMENT (RETURN_PEN-PPCL-T-0825).		
14	A1	48	P&ID NO. OF RO-23-2002 HAS BEEN REVISED IN ACCORDANCE WITH P&ID (RETURN_PEN-PPCL-T-0825).		
15	A1	10, 31	DESIGN TEMPERATURE OF FE/FT-23-1205 HAS BEEN REVISED IN ACCORDANCE WITH PIPE LINE CLASSIFICATION LIST.		
16	A1	10	DESIGN TEMPERATURE OF FXT-23-1204A HAS BEEN REVISED IN ACCORDANCE WITH PIPE LINE CLASSIFICATION LIST.		
17	A1	10	DESIGN TEMPERATURE OF FXT-23-1204B HAS BEEN REVISED IN ACCORDANCE WITH PIPE LINE CLASSIFICATION LIST.		
18	A1	10	DESIGN TEMPERATURE OF FXT-23-1204C HAS BEEN REVISED IN ACCORDANCE WITH PIPE LINE CLASSIFICATION LIST.		
19	F1	GENERAL	NOTE FOR PUMP MIN. FLOW HAS BEEN DELETED AS PER VENDOR INFORMATION.		
20	F1	INDEX, 46	PROCESS DATA OF FT-23-1004 HAS BEEN ADDED. INDEX HAS BEEN REVISED TO MEET CONSISTENT WITH DATASHEET.		
21	F1	GENERAL	STEAM OUT CONDITION OF THE ASSOCIATED FLOW INSTRUMENT HAVE BEEN REVISED AS PER PMT'S COMMENT		
			(RETURN_PEN-PPCL-T-0825).		
22	F1	29	LIQUID FLOW OF FE-23-0901 HAS BEEN REVISED FROM 34.5 M3/HR TO 34.8 M3/HR AS PER VENDOR INFORMATION.		
23	F1	49	FLOW RATE OF RO-23-2001 HAS BEEN REVISED FROM 6.72 M3/HR TO 3.3 M3/HR AS PER VENDOR INFORMATION.		
24	F2	9, 19	CALIBRATED RANGE OF FT-23-0401D HAS BEEN REVISED FROM 125 mBar TO 500 mBar AS PER PPCL REQUIREMENT.		
25	F2	47	RO-23-0501 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		
26	F2	47	RO-23-0701 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		
27	F2	47	RO-23-0801 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		
28	F2	48	RO-23-0901 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		
29	F2	48	NOTE 3 HAS BEEN SPECIFIED ON RO-23-1001 AS PER ITB REQUIREMENT.		
30	F2	49	NOTE 3 HAS BEEN ADDED AS PER ITB REQUIREMENT.		
31	F2	49	NOTE 3 HAS BEEN SPECIFIED ON RO-23-1203/2002 AS PER ITB REQUIREMENT.		
32	F2	49	RO-23-2001 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		
33	F2	49	RO-23-2201 DATA HAS BEEN REVISED AS A RESULT OF HYDRAULIC.		



PROJECT SPECIFICATION

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FLOW INSTRUMENT

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

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GENERAL REQUIREMENT AND INFORMATION FOR DIFFERENTIAL PRESSURE INSTRUMENT		8
FT-23- 0401D	Oxidate Feed to Preflash Column	9
FT-23-0403	Preflash Column Reflux	9
FT-23-0702	Preflash Column Bottoms to Flash Column Vaporizer	9
FT-23-0801	Flash Column Reflux	9
FT-23-0803	Preflash and Flash Columns Net Overhead Liquid	10
FT-23-0802	Preflash and Flash Columns Total Overhead Liquid	10
FT-23-0901	Flash Column Bottoms Pump Spillback	10
FT-23-1202	Decomposer Feed Startup Bypass	10
FT-23-1205	Flash Column Bottoms to Decomposer Drum	10
FXT-23- 1204A	Flash Column Bottoms to Decomposer Drum	10
FXT-23- 1204B	Flash Column Bottoms to Decomposer Drum	10
FXT-23- 1204C	Flash Column Bottoms to Decomposer Drum	10
FT-23-1301	Decomposer Product to Dehydrators	10
FT-23-1702	Decomposer Circulation Loop Elbow	10
FT-23-1801	Decomposer Circulation Liquid to Calorimeter No. 1	10
FXT-23-1803	Decomposer Circulation Liquid to Calorimeter No. 1	10
FT-23-1802	Decomposer Circulation Liquid to Calorimeter No. 2	10
FXT-23-1804	Decomposer Circulation Liquid to Calorimeter No. 2	10
FT-23-2001	Acid Aromatics to Direct Neutralizer Static Mixer	11
FT-23-0501	S1.5 Steam to Preflash Column Steam Heater	12
FT-23-0502	Preflash Column Steam Heater Condensate Pumps Discharge	12
FT-23-0701	S3 Steam to Flash Column Vaporizer	12
FT-23-0703	Flash Column Vaporizer Condensate Pumps Discharge	13
FT-23-1201	Process Water to Decomposer Feed	13
FT-23-1401	S4 Steam to Dehydrators	13
FT-23-1701	Reliable Cooling Water to Decomposer Cooler	13
FXT-23- 0401A	Oxidate Feed to Preflash Column	14
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FXT-23- 0401C	Oxidate Feed to Preflash Column	14
Note for Differential Pressure Instruments		15 - 17
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FE-23-0401	Oxidate Feed to Preflash Column	19
FE-23-0403	Preflash Column Reflux	20

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification:



PROJECT SPECIFICATION

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FLOW INSTRUMENT

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
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FE-23-0501	S1.5 Steam to Preflash Column Steam Heater	21
FE-23-0502	Preflash Column Steam Heater Condensate Pumps Discharge	22
FE-23-0701	S3 Steam to Flash Column Vaporizer	23
FE-23-0702	Preflash Column Bottoms to Flash Column Vaporizer	24
FE-23-0703	Flash Column Vaporizer Condensate Pumps Discharge	25
FE-23-0801	Flash Column Reflux	26
FE-23-0803	Preflash and Flash Columns Net Overhead Liquid	27
FE-23-0802	Preflash and Flash Columns Total Overhead Liquid	28
FE-23-0901	Flash Column Bottoms Pump Spillback	29
FE-23-1202	Decomposer Feed Startup Bypass	30
FE-23-1205	Flash Column Bottoms to Decomposer Drum	31
FE-23-1201	Process Water to Decomposer Feed	32
FE-23-1301	Decomposer Product to Dehydrators	33
FE-23-1401	S4 Steam to Dehydrators	34
FE-23-1701	Reliable Cooling Water to Decomposer Cooler	35
FE-23-1801, 1802	Decomposer Circulation Liquid to Calorimeter	36
FE-23-2001	Acid Aromatics to Direct Neutralizer Static Mixer	37

MASS FLOW METER

FT-23-1003	Sealant Return from Concentration Vacuum Producing Equipment	38
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FT-23-1903	Diamine Injection to Direct Neutralization Static Mixer	41

THERMAL MASS FLOW INSTRUMENT

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FXSL-23-1203	Cumene Flush to Oxidate Long Circulation Line	43
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RESTRICTION ORIFICE

RO-23-0501	Preflash Column Steam heater Condensate Pump Min. Flow Line	47
RO-23-0601	Cumene to Concentration Cumene Quench Drum	47
RO-23-0701	Flash Column Vaporizer Condensate Pump Min. Flow Line	47
RO-23-0801	Preflash And Flash Columns Overhead Pumps Start Up	47
RO-23-0901	Flash Column Bottoms Pump Spill Back	48
RO-23-1001	Nitrogen to Concentration Vacuum Producing Equipment	48
RO-23-1201	CHP Line Cumene Flush	48
RO-23-1202	CHP Line drain	48

Other specifications	UOP Standard Specification:
referenced in this	UOP Standard Drawing:
specification:	UOP Project Specification:



FLOW INSTRUMENT

PROJECT SPECIFICATION

SHEET 5 / 50

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
RO-23-1203	Process Water to Decomposer Drum	49
RO-23-2001	Acid Aromatics Sump Pumps Min. Flow	49
RO-23-2002	Nitrogen to Acid Aromatics Sump	49
RO-23-2201	Nitrogen to Acid Aromatics Sub header Purge	49
ROTAMETER		
FIF-23-2201	Nitrogen to Aromatics Relief Sub Header	50

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification:

Revision
Indication

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ENGINEERING

INSTRUMENT GENERAL REQUIREMENTS
PROJECT SPECIFICATION
963766 — **601**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

This specification shall be supplementary to all Project Specifications, 602 - 699, issued under this project number. Unless specified otherwise on the Project Specification, the following shall apply:

1 Climatic Ambient Temperature Range

The climatic ambient temperature range for field equipment design is 15 to 40°C.

2 Electrical Classification

The area electrical classification shall be established by the Licensee/Contractor in accordance with local electrical codes. The Contractor shall advise all equipment vendors of the area electrical classification.

3 Flanges

3.1 Carbon steel flanges are based on Material Group 1.1 of ASME B16.5 and ASME B16.47. Flange classes may differ when instrument flanges and the mating flanges on piping or equipment are specified with dissimilar materials. The Contractor shall verify that all mating flanges are properly matched with the specified instrument flanges.

3.2 Flange classes are specified in accordance with ASME B16.5 or ASME B16.47. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion, including joints between dissimilar materials, and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.

3.3 Flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum. Flanges intended for use with other gaskets shall have a flange finish within the optimal range for the specified gasket. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1. Flange finishes shall be protected from damage during fabrication, heat treatment, shipping, storage, and installation.

4 Tropical Environment

All equipment shall be tropicalized to eliminate mildew, fungi, and other detrimental effects of a tropical environment. Packaging shall be suitable for shipment and storage under tropical conditions. Packaging and "tropicalization" subject to approval by the licensee.

5 Marine Environment

All field equipment shall be suitable for operation in a corrosive, salt laden, marine atmosphere.

6 Tropical and Marine Environment

Transmitters and all equipment/instruments containing electrical parts shall be installed under rain shield.

Revision
Indication

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INSTRUMENT GENERAL REQUIREMENTS

PROJECT SPECIFICATION

963766 — 601

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

7 Fieldbus Transmitters

This specification is based on conventional (4 – 20 mA output) transmitters. UOP does not require the use of fieldbus transmitters and, if installed incorrectly, the fieldbus system can be detrimental to the control and operation of the unit. The implementation of these fieldbus transmitters is the Contractor's responsibility.

Revision
Indication

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-G

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

6-115	X	6-116	6-120	6-122 + 6-123	6-139	6-140
6-142	X	6-144	6-147	6-148	6-159	6-161
						6-162

2 Sensor Range

The Sensing Element Lower Range Limit and Upper Range Limit are to be used as a guide only. Transmitters that have greater range capabilities are acceptable.

3 Output Indicator

When an Output Indicator is specified, an integrally mounted indicator is acceptable, provided the indicator is readable from the associated control valve assembly (or other valve as denoted on the P&ID). If the piping layout is such that an integrally mounted indicator is not readable from the associated valve, then a separate indicator suitable for remote mounting shall be supplied.

4 Process Temperature

The temperature shown is the *process* temperature and is not normally the operating temperature of the transmitter.

Other specifications
referenced in this
specification:

UOP Standard Specification: 6-18
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 1 Quantity: 21 Data Page: 1 of 3										
BODY	Body/Flange Material: 316 stainless steel ‡ Static Pressure Rating: 105 kg/cm2(g) Process Connections Size: 1/2 inch Process Connections Material: 316 stainless steel ‡ Drains/Vents Material: 316 stainless steel ‡ Gasket/O-Ring Material: Man. Recommendation Gasket Type: Manufacturer's Recommendation Bolting/Nut Material: Manufacturer's Recommendation	HIGH SIDE REMOTE SEAL	Process Conn Size: Type: Class: Flange Facing: Flange Face Finish: Seal Type: Fill Fluid: Diaphragm Material: Mounting Flange Material: Non-Wetted Housing Material: Capillary Length: Armor Material: Diaphragm Extension Length:							
	Diaphragm/Wetted Material: 316 stainless steel ‡ Non-Wetted Material: Manufacturer's Recommendation Zero Elevation/Suppression: Provide Sensor Lower Range Limit: 60 mBar Sensor Upper Range Limit: 370 mBar Fill Fluid: Silicone		LOW SIDE REMOTE SEAL	Process Conn Size: Type: Class: Flange Facing: Flange Face Finish: Seal Type: Fill Fluid: Diaphragm Material: Mounting Flange Material: Non-Wetted Housing Material: Capillary Length: Armor Material: Diaphragm Extension Length:						
	Transmitter Output: 4 - 20 mA Adjustable Dampening: Housing Material: Manufacturer's Recommendation Manufacturer: Honeywell, Rosemount (or equal)			ACCESSORIES	Manifold Type: Manifold Material: Manifold Manufacturer: Output Indicator: See Remark § Air Filter/Regulator Quantity:					
	Function: Controller Type: Controller Action: Controller Output: Manufacturer:									
	§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential. ‡ All transmitter parts in contact with the process shall be 316 stainless steel. DP : Design Pressure DT : Design Temperature									
	Tag Number				Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	FT-23-0401D				Oxidate Feed to Preflash Column	0 to 500 mBar	82 °C	7.0 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 11.5 kg/cm2(g), F.V / DT : 120 degC
	FT-23-0403				Preflash Column Reflux	0 to 250 mBar	39 °C	7.2 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 13.55 kg/cm2(g), F.V / DT : 120 degC
	FT-23-0702				Preflash Column Bottoms to Flash Column Vaporizer	0 to 125 mBar	70 °C	-0.6 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 4.0 kg/cm2(g), F.V / DT : 250 degC
	FT-23-0801				Flash Column Reflux	0 to 250 mBar	39 °C	7.2 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 13.55 kg/cm2(g), F.V / DT : 120 degC
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.										

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PROJECT SPECIFICATION

963766 — 604

DIFFERENTIAL PRESSURE INSTRUMENTS

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 1

REV	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	FT-23-0803	Preflash and Flash Columns Net Overhead Liquid	0 to 250 mBar	39 °C	7.2 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 13.55 kg/cm2(g), F.V / DT : 120 degC
	FT-23-0802	Preflash and Flash Columns Total Overhead Liquid	0 to 250 mBar	39 °C	7.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 13.55 kg/cm2(g), F.V / DT : 120 degC
	FT-23-0901	Flash Column Bottoms Pump Spillback	0 to 250 mBar	60 °C	3.5 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 8.86 kg/cm2(g), F.V / DT : 250 degC
	FT-23-1202	Decomposer Feed Startup Bypass	0 to 250 mBar	60 °C	3.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 10.0 kg/cm2(g), F.V / DT : 250 degC
	FT-23-1205	Flash Column Bottoms to Decomposer Drum	0 to 125 mBar	60 °C	3.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 8.66 kg/cm2(g), F.V / DT : 195 degC
	FXT-23-1204A	Flash Column Bottoms to Decomposer Drum	0 to 125 mBar	60 °C	3.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 8.66 kg/cm2(g), F.V / DT : 195 degC
	FXT-23-1204B	Flash Column Bottoms to Decomposer Drum	0 to 125 mBar	60 °C	3.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 8.66 kg/cm2(g), F.V / DT : 195 degC
	FXT-23-1204C	Flash Column Bottoms to Decomposer Drum	0 to 125 mBar	60 °C	3.4 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 8.66 kg/cm2(g), F.V / DT : 195 degC
	FT-23-1301	Decomposer Product to Dehydrators	0 to 250 mBar	83 °C	9.0 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. DP : 21.0 kg/cm2(g), F.V / DT : 250 degC
	FT-23-1702	Decomposer Circulation Loop Elbow	0 to 293 mBar	58 °C	2.7 kg/cm2(g)	Note 14	Impulse piping and fittings, shall be 316 stainless steel. Notes 13, 14 DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	FT-23-1801	Decomposer Circulation Liquid to Calorimeter No. 1	0 to 250 mBar	79 °C	3.9 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 15 DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	FXT-23-1803	Decomposer Circulation Liquid to Calorimeter No. 1	0 to 250 mBar	79 °C	3.9 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 15 DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	FT-23-1802	Decomposer Circulation Liquid to Calorimeter No. 2	0 to 250 mBar	79 °C	3.9 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 15 DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	FXT-23-1804	Decomposer Circulation Liquid to Calorimeter No. 2	0 to 250 mBar	79 °C	3.9 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 15 DP : 16.5 kg/cm2(g), F.V / DT : 250 degC

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DISTRIBUTION (continued)

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PROJECT SPECIFICATION

963766 — 604

DIFFERENTIAL PRESSURE INSTRUMENTS

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

1		Item Number: 4		Quantity: 11		Data Page: 1 of 2	
BODY	Body/Flange Material: Cadmium plated CS or 316ss			HIGH SIDE REMOTE SEAL	Process Conn Size: _____ Type: _____ Class: _____		
	Static Pressure Rating: 105 kg/cm2(g)				Flange Facing: _____ Flange Face Finish: _____		
	Process Connections Size: 1/2 inch				Seal Type: _____		
	Process Connections Material: Cadmium plated CS or 316ss				Fill Fluid: _____		
	Drains/Vents Material: 316 stainless steel				Diaphragm Material: _____		
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material: _____		
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material: _____		
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: _____ Armor Material: _____		
	Diaphragm/Wetted Material: 316 stainless steel				Diaphragm Extension Length: _____		
	Non-Wetted Material: Manufacturer's Recommendation				LOW SIDE REMOTE SEAL	Process Conn Size: _____ Type: _____ Class: _____	
Zero Elevation/Suppression: Provide			Flange Facing: _____ Flange Face Finish: _____				
Sensor Lower Range Limit: 60 mBar			Seal Type: _____				
Sensor Upper Range Limit: 370 mBar			Fill Fluid: _____				
Fill Fluid: Silicone			Diaphragm Material: _____				
Transmitter Output: 4 - 20 mA			Mounting Flange Material: _____				
Adjustable Dampening: _____			Non-Wetted Housing Material: _____				
Housing Material: Manufacturer's Recommendation			Capillary Length: _____ Armor Material: _____				
Manufacturer: Honeywell, Rosemount (or equal)			Diaphragm Extension Length: _____				
INDICATOR / CONTROLLER	Function: _____			ACCESSORIES		Manifold Type: _____	
	Controller Type: _____				Manifold Material: _____		
	Controller Action: _____				Manifold Manufacturer: _____		
	Controller Output: _____				Output Indicator: See Remark §		
	Manufacturer: _____				Air Filter/Regulator Quantity: _____		
REMARKS	§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.						
	DP : Design Pressure						
	DT : Design Temperature						
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	FT-23-0501	S1.5 Steam to Preflash Column Steam Heater	0 to 62.5 mBar	135 °C	1.5 kg/cm2(g)	6-116	DP : 7.0 kg/cm2(g), F.V / DT : 195 degC
	FT-23-0502	Preflash Column Steam Heater Condensate Pumps Discharge	0 to 250 mBar	117 °C	2.8 kg/cm2(g)	6-116	DP : 9.233 kg/cm2(g), F.V / DT : 195 degC
	FT-23-0701	S3 Steam to Flash Column Vaporizer	0 to 125 mBar	148 °C	2.7 kg/cm2(g)	6-116	DP : 10.5 kg/cm2(g), F.V / DT : 195 degC
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.							

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[illegible]



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 7 Quantity: 4 Data Page: 1 of 1										
BODY	Body/Flange Material: 316 stainless steel ‡ Static Pressure Rating: 105 kg/cm2(g) Process Connections Size: 1/2 inch Process Connections Material: 316 stainless steel ‡ Drains/Vents Material: 316 stainless steel ‡ Gasket/O-Ring Material: Man. Recommendation Gasket Type: Manufacturer's Recommendation Bolting/Nut Material: Manufacturer's Recommendation	HIGH SIDE REMOTE SEAL	Process Conn Size: Type: Class: Flange Facing: Flange Face Finish: Seal Type: Fill Fluid: Diaphragm Material: Mounting Flange Material: Non-Wetted Housing Material: Capillary Length: Armor Material: Diaphragm Extension Length:							
	Diaphragm/Wetted Material: 316 stainless steel ‡ Non-Wetted Material: Manufacturer's Recommendation Zero Elevation/Suppression: Provide Sensor Lower Range Limit: 300 mBar Sensor Upper Range Limit: 1860 mBar Fill Fluid: Silicone		LOW SIDE REMOTE SEAL	Process Conn Size: Type: Class: Flange Facing: Flange Face Finish: Seal Type: Fill Fluid: Diaphragm Material: Mounting Flange Material: Non-Wetted Housing Material: Capillary Length: Armor Material: Diaphragm Extension Length:						
	Transmitter Output: 4 - 20 mA Adjustable Dampening: Housing Material: Manufacturer's Recommendation Manufacturer: Honeywell, Rosemount (or equal)			ACCESSORIES	Manifold Type: Manifold Material: Manifold Manufacturer: Output Indicator: See Remark § Air Filter/Regulator Quantity:					
	Function: Controller Type: Controller Action: Controller Output: Manufacturer:									
	§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential. ‡ All transmitter parts in contact with the process shall be 316 stainless steel. DP : Design Pressure DT : Design Temperature									
	Tag Number Service Calibrated Range Process Temperature Process Pressure Installation Comments									
	FXT -23-0401A				Oxidate Feed to Preflash Column	0 to 500 mBar	82 °C	7.0 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 11.5 kg/cm2(g), F.V / DT : 120 degC
	FXT -23-0401B				Oxidate Feed to Preflash Column	0 to 500 mBar	82 °C	7.0 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 11.5 kg/cm2(g), F.V / DT : 120 degC
	FXT -23-0401C				Oxidate Feed to Preflash Column	0 to 500 mBar	82 °C	7.0 kg/cm2(g)	6-116	Impulse piping and fittings, shall be 316 stainless steel. Note 1 DP : 11.5 kg/cm2(g), F.V / DT : 120 degC

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

Revision
Indication

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A Honeywell Company


DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Notes:

- Four transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
- The transmitter differential range specified is based upon a level range of ~~1600~~ 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- Impulse piping and fittings shall be 316 stainless steel.
- The transmitter differential range specified is based upon a level range of 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.947. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.931. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 2400 mm and a liquid specific gravity of 0.935. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 4200 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Notes:

10. The transmitter differential range specified is based upon a level range of 1400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
11. The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
12. The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
13. Provide a first order lag signal filtering function, with adjustable time constant, for reduction of process signal noise. The signal filtering may be accomplished either by analog or digital means, at the transmitter. Alternatively, the signal filtering can be performed at the DCS. The time constant shall initially be set at the minimum value.
14. A horizontal long radius elbow in the regular piping shall be used for this meter. The pressure taps shall be located at an angle of 45 degrees from the inlet weld seam, diametrically opposite one another, at the extreme inside and outside of the bend. Provide 25 diameters of straight run pipe upstream of the long radius elbow and 10 diameters downstream. The meter manifold and meter installation shall otherwise conform to UOP Standard Drawing 6-116.
15. Two transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
16. The transmitter differential range specified is based upon a level range of 2100 mm and a liquid specific gravity of 0.800. Fill the reference leg with Acid Aromatics Knockout Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
17. The transmitter differential range specified is based upon a level range of 2500 mm and a liquid specific gravity of 0.848. Fill the reference leg with Preflash and Flash Columns Condenser liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

Revision
Indication

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A Honeywell Company



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Notes:

18. The transmitter differential range specified is based upon a level range of 1000 mm and a liquid specific gravity of 1.716. Provide zero elevation as required to compensate for the static head of capillary fill fluid. ~~between the upper and lower level connections and the transmitter.~~ The low pressure tap shall be vented to the atmosphere and protected from rain, wind and insects. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
19. Contractor to confirm the calibration range.

Revision
Indication

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**PROJECT SPECIFICATION**963766 — **605-G**
**PRIMARY FLOW ELEMENTS
(DIFFERENTIAL PRESSURE TYPE)**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION**1 Standard Drawings Referenced**

UOP Standard Drawings as indicated below are part of this specification.

X	6-111	X	6-112		6-134						
---	-------	---	-------	--	-------	--	--	--	--	--	--

2 Standard Specification Referenced

UOP Standard Specification 6-18 is part of this specification.

3 Sizing Calculations

Primary element sizing calculations shall be provided for each primary element, and one copy shall be retained by the contractor for review by UOP startup personnel.

4 Surface Finish

When raised face flanges are indicated, the gasket contact portion of the orifice plate shall have a surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum, except as noted.

 Other specifications
referenced in this
specification:

 UOP Standard Specification: 6-18
 UOP Standard Drawing: See item 1 above
 UOP Project Specification: 601, this project



PROJECT SPECIFICATION

963766 — 605

PRIMARY FLOW ELEMENTS
(DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **1** Quantity: **1** Tag Number: **FE-23-0401**
 Service: **Oxidate Feed to Preflash Column**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit
		Meter Maximum at BASE Condition		260	m3/h	Vapor Specific Gravity (Air=1)	
	Differential Press Inst Calibration		500	mBar	Temperature	82	°C
	Liquid Flow at BASE Temperature		225.9	m3/h	Inlet Pressure	7.0	kg/cm2(g)
	Vapor Flow at Base Condition		normal m3/h		Liquid Viscosity	0.607	cP
	Steam Flow		kg/h		Vapor Viscosity		cP
	Liq Specific Gravity at BASE Temp		0.906	----	Specific Heats Cp/Cv (Real)		
	Liq Specific Gravity at Flow Temp		0.846	----	Compressibility Z		

ORIFICE PLATE	ORIFICE FLANGE		PREFABRICATED METER RUN	VENTURI	INSERTION	INSTAL-LATION	BASE	SPECIAL
	Bore Type: Eccentric § Bore Inlet Edge Style: Square Bore Outlet Style: Orifice Bore Diameter: Beta Ratio (d/D): Orifice Plate Material: 304 or 316 Stainless Steel Plate Thickness: Plate Outside Diameter: Vent/Drain Hole Size: Vent/Drain Location: Gasket Contact Face Finish: Holding Ring Type: Holding Ring Material: Orifice Plate Drawing: §	End Conn Size: 8" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Pressure Tap Conn Size: 1/2" Type: NPT Pressure Tap Style: Flange Gasket Style: Pressure Tap Orientation: Meter Run Nominal Size:						

REMARKS
§ If the orifice plate is located in a horizontal meter run, the orifice plate shall be fabricated in accordance with UOP Standard Drawing 6-111, Detail 'A', except that the orifice plate shall be the eccentric type. If reducers are used in the horizontal meter run they shall be eccentric and installed flat side down, and the upstream straight meter run shall be in accordance with UOP Standard Drawing 6-112. For startup operation, FT-23-0401D will be calibrated for 500 mBar with a flowrate of 53.1 m3/h. Provide a Meter Maximum flow calculation for this operation. Design Pressure : 11.5 kg/cm2(g) / F.V Design Temperature : 120 degC

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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PROJECT SPECIFICATION

963766 — 605

PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **2** Quantity: **1** Tag Number: **FE-23-0403**
Service: **Preflash Column Reflux**

SIZING INFORMATION		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	30	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	250	mBar	Temperature	39 °C
		Liquid Flow at BASE Temperature	22.4	m3/h	Inlet Pressure	7.2 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	0.622 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	0.867	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.848	----	Compressibility Z	
ORIFICE PLATE		Bore Type: Concentric			Venturi Type:	
		Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:	
		Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:	
		Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)	
		Gasket Contact Face Finish:			Pressure Tap Orientation:	
		Holding Ring Type:			Insertion Type:	
		Holding Ring Material:			Sensor Configuration:	
ORIFICE FLANGE		Orifice Plate Drawing: 6-111			Sensor Conn Size: Type: Class:	
		End Conn Size: 3" Type: Flanged Class: 300			Flange Facing: Flange Face Finish:	
		Flange Facing: Raised Face Flange Face Finish:			Sensor/Conn Material:	
		Pressure Tap Conn Size: 1/2" Type: NPT			Sensor/Conn Style:	
		Pressure Tap Style: Flange			Other Wetted Components Material:	
		Gasket Style:			Inst Conn Size: Inst Conn Mat:	
		Pressure Tap Orientation:			Inst Valve Type:	
		Meter Run Nominal Size:			Accuracy:	
PREFABRICATED METER RUN		End Conn Size: Type: Class:			Repeatability:	
		Orifice Flange Conn Size: Class:			Model Number:	
		Flange Facing: Flange Face Finish:			Manufacturer:	
		Pressure Tap Style:			Pipe Nominal Size: 3"	
		Pressure Tap Conn Size: Type:			Estimated Inside Pipe Diameter:	
		Pipe Schedule: Pipe Bore Diameter:			Straight Run Requirement: UOP Std. Dwg. 6-112	
		Upstream Length: Downstream Length:			Liquid-Base Temperature: 15°C	
		Pipe/Orifice Flange/End Conn Material:			Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Bolting/Nut Material:			Calculation Method:	
		Gasket Style: Gasket Material:			Primary Flow Element Calibration:	
REMARKS		Design Pressure : 13.55 kg/cm2(g) / F.V				
		Design Temperature : 120 degC				

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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PROJECT SPECIFICATION

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **3** Quantity: **1** Tag Number: **FE-23-0501**
 Service: **S1.5 Steam to Preflash Column Steam Heater**

SIZING INFORMATION	Process Data Case		Design	Unit	Sizing Case		Unit
	Meter Maximum at BASE Condition		15000	kg/h	Vapor Specific Gravity (Air=1)		0.622
	Differential Press Inst Calibration		62.5	mBar	Temperature		135
	Liquid Flow at BASE Temperature			m3/h	Inlet Pressure		1.5
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity		
	Steam Flow		11388	kg/h	Vapor Viscosity		0.014
	Liq Specific Gravity at BASE Temp			- - - -	Specific Heats Cp/Cv (Real)		1.303
	Liq Specific Gravity at Flow Temp			- - - -	Compressibility Z		0.974
	ORIFICE PLATE	Bore Type: Concentric			VENTURI	Venturi Type:	
Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:				
Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:				
Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:				
Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:				
Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:				
Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)				
Gasket Contact Face Finish:			Pressure Tap Orientation:				
Holding Ring Type:			Insertion Type:				
Holding Ring Material:			Sensor Configuration:				
ORIFICE FLANGE	Orifice Plate Drawing: 6-111			INSERTION	Sensor Conn Size: Type: Class:		
	End Conn Size: 18" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:		
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:		
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:		
	Pressure Tap Style: Flange				Other Wetted Components Material:		
	Gasket Style:				Inst Conn Size: Inst Conn Mat:		
	Pressure Tap Orientation:				Inst Valve Type:		
	Meter Run Nominal Size:				Accuracy:		
PREFABRICATED METER RUN	End Conn Size: Type: Class:			INSTAL- LATION	Repeatability:		
	Orifice Flange Conn Size: Class:				Model Number:		
	Flange Facing: Flange Face Finish:				Manufacturer:		
	Pressure Tap Style:				Pipe Nominal Size: 18"		
	Pressure Tap Conn Size: Type:			SPECIAL	Estimated Inside Pipe Diameter:		
	Pipe Schedule: Pipe Bore Diameter:				Straight Run Requirement: UOP Std. Dwg. 6-112		
	Upstream Length: Downstream Length:				Liquid-Base Temperature: 15°C		
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C		
	Bolting/Nut Material:				Calculation Method:		
	Gasket Style: Gasket Material:				Primary Flow Element Calibration:		
	Pressure Tap Orientation:						
	Manufacturer:						
REMARKS	Design Pressure : 7.0 kg/cm2(g) / F.V						
	Design Temperature : 195 degC						

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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 PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)				PROJECT SPECIFICATION 963766 — 605 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																												
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																								
0	25JUN12	MVD	MW																																												
Item Number 5 Quantity: 1 Tag Number: FE-23-0701 Service: S3 Steam to Flash Column Vaporizer																																															
SIZING INFORMATION	Process Data Case		Design	Unit	Sizing Case		Unit																																								
	Meter Maximum at BASE Condition		10500	kg/h	Vapor Specific Gravity (Air=1)		0.622																																								
	Differential Press Inst Calibration		125	mBar	Temperature		148																																								
	Liquid Flow at BASE Temperature			m3/h	Inlet Pressure		2.7																																								
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity																																										
	Steam Flow		7947	kg/h	Vapor Viscosity		0.014																																								
	Liq Specific Gravity at BASE Temp			----	Specific Heats Cp/Cv (Real)		1.301																																								
	Liq Specific Gravity at Flow Temp			----	Compressibility Z		0.967																																								
ORIFICE PLATE	Bore Type: Concentric			VENTURI	Venturi Type:																																										
	Bore Inlet Edge Style: Square				Venturi Conn Size: Type: Class:																																										
	Bore Outlet Style: Bevel				Flange Facing: Flange Face Finish:																																										
	Orifice Bore Diameter: Beta Ratio (d/D):				Venturi Material:																																										
	Orifice Plate Material: 304 or 316 Stainless Steel				Venturi Wall Thickness:																																										
	Plate Thickness: Plate Outside Diameter:				Pressure Tap Conn Size: Type:																																										
	Vent/Drain Hole Size: Vent/Drain Location:				Beta Ratio (d/D)																																										
	Gasket Contact Face Finish:				Pressure Tap Orientation:																																										
	Holding Ring Type:				Insertion Type:																																										
	Holding Ring Material:				Sensor Configuration:																																										
ORIFICE FLANGE	Orifice Plate Drawing: 6-111			INSERTION	Sensor Conn Size: Type: Class:																																										
	End Conn Size: 12" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:																																										
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:																																										
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:																																										
	Pressure Tap Style: Flange				Other Wetted Components Material:																																										
	Gasket Style:				Inst Conn Size: Inst Conn Mat:																																										
	Pressure Tap Orientation:				Inst Valve Type:																																										
	Meter Run Nominal Size:				Accuracy:																																										
PREFABRICATED METER RUN	End Conn Size: Type: Class:			INSTAL-LATION	Repeatability:																																										
	Orifice Flange Conn Size: Class:				Model Number:																																										
	Flange Facing: Flange Face Finish:				Manufacturer:																																										
	Pressure Tap Style:				Pipe Nominal Size: 12"																																										
	Pressure Tap Conn Size: Type:				Estimated Inside Pipe Diameter:																																										
	Pipe Schedule: Pipe Bore Diameter:				Straight Run Requirement: UOP Std. Dwg. 6-112																																										
	Upstream Length: Downstream Length:				Liquid-Base Temperature: 15°C																																										
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C																																										
	Bolting/Nut Material:				Calculation Method:																																										
	Gasket Style: Gasket Material:				Primary Flow Element Calibration:																																										
REMARKS	Pressure Tap Orientation:			SPECIAL																																											
	Manufacturer:																																														
	Design Pressure : 10.5 kg/cm2(g) / F.V																																														
	Design Temperature : 195 degC																																														

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.



PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

PROJECT SPECIFICATION

963766 — 605

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **6** Quantity: **1** Tag Number: **FE-23-0702**

Service: **Preflash Column Bottoms to Flash Column Vaporizer**

Sizing Information		Process Data Case		Single Case	Unit	Sizing Case		Unit
SIZING INFORMATION		Meter Maximum at BASE Condition		120	m3/h	Vapor Specific Gravity (Air=1)		----
		Differential Press Inst Calibration		125	mBar	Temperature		70 °C
		Liquid Flow at BASE Temperature		88.0	m3/h	Inlet Pressure		-0.61 kg/cm2(g)
		Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity		1.507 cP
		Steam Flow			kg/h	Vapor Viscosity		cP
		Liq Specific Gravity at BASE Temp		0.969	----	Specific Heats Cp/Cv (Real)		
		Liq Specific Gravity at Flow Temp		0.916	----	Compressibility Z		
ORIFICE PLATE		Bore Type: Eccentric §				VENTURI	Venturi Type:	
		Bore Inlet Edge Style: Square					Venturi Conn Size: Type: Class:	
		Bore Outlet Style:					Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):					Venturi Material:	
		Orifice Plate Material: 304 or 316 Stainless Steel					Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:					Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:					Beta Ratio (d/D)	
		Gasket Contact Face Finish:					Pressure Tap Orientation:	
		Holding Ring Type:					Insertion Type:	
		Holding Ring Material:						
ORIFICE FLANGE		End Conn Size: 8" Type: Flanged Class: 300				INSERTION	Sensor Configuration:	
		Flange Facing: Raised Face Flange Face Finish:					Sensor Conn Size: Type: Class:	
		Pressure Tap Conn Size: 1/2" Type: NPT					Flange Facing: Flange Face Finish:	
		Pressure Tap Style: Flange					Sensor/Conn Material:	
		Gasket Style:					Sensor/Conn Style:	
		Pressure Tap Orientation:					Other Wetted Components Material:	
		Meter Run Nominal Size:					Inst Conn Size: Inst Conn Mat:	
							Inst Valve Type:	
PREFABRICATED METER RUN		End Conn Size: Type: Class:				INSTAL-LATION	Accuracy:	
		Orifice Flange Conn Size: Class:					Repeatability:	
		Flange Facing: Flange Face Finish:					Model Number:	
		Pressure Tap Style:					Manufacturer:	
		Pressure Tap Conn Size: Type:				BASE	Pipe Nominal Size: 8"	
		Pipe Schedule: Pipe Bore Diameter:					Estimated Inside Pipe Diameter:	
		Upstream Length: Downstream Length:					Straight Run Requirement: §	
		Pipe/Orifice Flange/End Conn Material:				SPECIAL	Liquid-Base Temperature: 15°C	
		Bolting/Nut Material:					Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Gasket Style: Gasket Material:					Calculation Method:	
	Pressure Tap Orientation:				Primary Flow Element Calibration:			
	Manufacturer:							
REMARKS		§ If the orifice plate is located in a horizontal meter run, the orifice plate shall be fabricated in accordance with UOP Standard Drawing 6-111, Detail 'A', except that the orifice plate shall be the eccentric type. If reducers are used in the horizontal meter run they shall be eccentric and installed flat side down, and the upstream straight meter run shall be in accordance with UOP Standard Drawing 6-112.						
		Design Pressure : 4.0 kg/cm2(g) / F.V						
		Design Temperature : 250 degC						

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **7** Quantity: **1** Tag Number: **FE-23-0703**Service: **Flash Column Vaporizer Condensate Pumps Discharge**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit
	Meter Maximum at BASE Condition		15	m3/h	Vapor Specific Gravity (Air=1)		----
	Differential Press Inst Calibration		250	mBar	Temperature		135 °C
	Liquid Flow at BASE Temperature		11.6	m3/h	Inlet Pressure		4.1 kg/cm2(g)
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity		0.204 cP
	Steam Flow			kg/h	Vapor Viscosity		cP
	Liq Specific Gravity at BASE Temp		1.000	----	Specific Heats Cp/Cv (Real)		
	Liq Specific Gravity at Flow Temp		0.931	----	Compressibility Z		
	ORIFICE PLATE	Bore Type: Concentric			VENTURI	Venturi Type:	
Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:				
Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:				
Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:				
Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:				
Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:				
Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)				
Gasket Contact Face Finish:			Pressure Tap Orientation:				
Holding Ring Type:			Insertion Type:				
Holding Ring Material:			Sensor Configuration:				
ORIFICE FLANGE	Orifice Plate Drawing: 6-111			INSERTION	Sensor Conn Size: Type: Class:		
	End Conn Size: 3" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:		
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:		
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:		
	Pressure Tap Style: Flange				Other Wetted Components Material:		
	Gasket Style:				Inst Conn Size: Inst Conn Mat:		
	Pressure Tap Orientation:				Inst Valve Type:		
	Meter Run Nominal Size:				Accuracy:		
PREFABRICATED METER RUN	End Conn Size: Type: Class:			INSTAL- LATION	Repeatability:		
	Orifice Flange Conn Size: Class:				Model Number:		
	Flange Facing: Flange Face Finish:				Manufacturer:		
	Pressure Tap Style:				Pipe Nominal Size: 3"		
	Pressure Tap Conn Size: Type:			SPECIAL	Estimated Inside Pipe Diameter:		
	Pipe Schedule: Pipe Bore Diameter:				Straight Run Requirement: UOP Std. Dwg. 6-112		
	Upstream Length: Downstream Length:				Liquid-Base Temperature: 15°C		
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C		
	Bolting/Nut Material:				Calculation Method:		
	Gasket Style: Gasket Material:				Primary Flow Element Calibration:		
	Pressure Tap Orientation:						
	Manufacturer:						
REMARKS	Design Pressure : 9.233 kg/cm2(g) / F.V						
	Design Temperature : 195 degC						

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PRIMARY FLOW ELEMENTS
(DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **8** Quantity: **1** Tag Number: **FE-23-0801**
 Service: **Flash Column Reflux**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit
	Meter Maximum at BASE Condition		28	m3/h	Vapor Specific Gravity (Air=1)		----
	Differential Press Inst Calibration		250	mBar	Temperature		39 °C
	Liquid Flow at BASE Temperature		21.4	m3/h	Inlet Pressure		7.2 kg/cm2(g)
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity		0.622 cP
	Steam Flow			kg/h	Vapor Viscosity		cP
	Liq Specific Gravity at BASE Temp		0.867	----	Specific Heats Cp/Cv (Real)		
	Liq Specific Gravity at Flow Temp		0.848	----	Compressibility Z		
	ORIFICE PLATE	Bore Type: Concentric			VENTURI	Venturi Type:	
Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:				
Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:				
Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:				
Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:				
Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:				
Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)				
Gasket Contact Face Finish:			Pressure Tap Orientation:				
Holding Ring Type:			Insertion Type:				
Holding Ring Material:			Sensor Configuration:				
ORIFICE FLANGE	Orifice Plate Drawing: 6-111			INSERTION	Sensor Conn Size: Type: Class:		
	End Conn Size: 3" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:		
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:		
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:		
	Pressure Tap Style: Flange				Other Wetted Components Material:		
	Gasket Style:				Inst Conn Size: Inst Conn Mat:		
	Pressure Tap Orientation:				Inst Valve Type:		
	Meter Run Nominal Size:				Accuracy:		
PREFABRICATED METER RUN	End Conn Size: Type: Class:			INSTAL- LATION	Repeatability:		
	Orifice Flange Conn Size: Class:				Model Number:		
	Flange Facing: Flange Face Finish:				Manufacturer:		
	Pressure Tap Style:				Pipe Nominal Size: 3"		
	Pressure Tap Conn Size: Type:			SPECIAL BASE	Estimated Inside Pipe Diameter:		
	Pipe Schedule: Pipe Bore Diameter:				Straight Run Requirement: UOP Std. Dwg. 6-112		
	Upstream Length: Downstream Length:				Liquid-Base Temperature: 15°C		
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C		
	Bolting/Nut Material:				Calculation Method:		
	Gasket Style: Gasket Material:				Primary Flow Element Calibration:		
	Pressure Tap Orientation:						
	Manufacturer:						
REMARKS	Design Pressure : 13.55 kg/cm2(g) / F.V						
	Design Temperature : 120 degC						

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **9** Quantity: **1** Tag Number: **FE-23-0803**Service: **Preflash and Flash Columns Net Overhead Liquid**

SIZING INFORMATION		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	215	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	250	mBar	Temperature	39 °C
		Liquid Flow at BASE Temperature	162.6	m3/h	Inlet Pressure	7.2 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	0.622 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	0.867	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.848	----	Compressibility Z	
ORIFICE PLATE		Bore Type: Concentric			Venturi Type:	
		Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:	
		Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:	
		Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)	
		Gasket Contact Face Finish:			Pressure Tap Orientation:	
		Holding Ring Type:			Insertion Type:	
		Holding Ring Material:			Sensor Configuration:	
ORIFICE FLANGE		Orifice Plate Drawing: 6-111			Sensor Conn Size: Type: Class:	
		End Conn Size: 8" Type: Flanged Class: 300			Flange Facing: Flange Face Finish:	
		Flange Facing: Raised Face Flange Face Finish:			Sensor/Conn Material:	
		Pressure Tap Conn Size: 1/2" Type: NPT			Sensor/Conn Style:	
		Pressure Tap Style: Flange			Other Wetted Components Material:	
		Gasket Style:			Inst Conn Size: Inst Conn Mat:	
		Pressure Tap Orientation:			Inst Valve Type:	
		Meter Run Nominal Size:			Accuracy:	
PREFABRICATED METER RUN		End Conn Size: Type: Class:			Repeatability:	
		Orifice Flange Conn Size: Class:			Model Number:	
		Flange Facing: Flange Face Finish:			Manufacturer:	
		Pressure Tap Style:			Pipe Nominal Size: 8"	
		Pressure Tap Conn Size: Type:			Estimated Inside Pipe Diameter:	
		Pipe Schedule: Pipe Bore Diameter:			Straight Run Requirement: UOP Std. Dwg. 6-112	
		Upstream Length: Downstream Length:			Liquid-Base Temperature: 15°C	
		Pipe/Orifice Flange/End Conn Material:			Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Bolting/Nut Material:			Calculation Method:	
		Gasket Style: Gasket Material:			Primary Flow Element Calibration:	
REMARKS		Design Pressure : 13.55 kg/cm2(g) / F.V				
		Design Temperature : 120 degC				

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **10** Quantity: **1** Tag Number: **FE-23-0802**Service: **Preflash and Flash Columns Total Overhead Liquid**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit
		Meter Maximum at BASE Condition	275	m3/h	Vapor Specific Gravity (Air=1)		----
	Differential Press Inst Calibration	250	mBar	Temperature		39	
	Liquid Flow at BASE Temperature	206.4	m3/h	Inlet Pressure		7.4	
	Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity		0.622	
	Steam Flow		kg/h	Vapor Viscosity		cP	
	Liq Specific Gravity at BASE Temp	0.867	----	Specific Heats Cp/Cv (Real)			
	Liq Specific Gravity at Flow Temp	0.848	----	Compressibility Z			

ORIFICE PLATE	ORIFICE FLANGE		PREFABRICATED METER RUN	VENTURI	INSERTION	INSTAL-LATION	BASE	SPECIAL
	Bore Type: Concentric Bore Inlet Edge Style: Square Bore Outlet Style: Bevel Orifice Bore Diameter: Beta Ratio (d/D): Orifice Plate Material: 304 or 316 Stainless Steel Plate Thickness: Plate Outside Diameter: Vent/Drain Hole Size: Vent/Drain Location: Gasket Contact Face Finish: Holding Ring Type: Holding Ring Material: Orifice Plate Drawing: 6-111	End Conn Size: 8" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Pressure Tap Conn Size: 1/2" Type: NPT Pressure Tap Style: Flange Gasket Style: Pressure Tap Orientation: Meter Run Nominal Size:						

REMARKS

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **11** Quantity: **1** Tag Number: **FE-23-0901**Service: **Flash Column Bottoms Pump Spillback**

SIZING INFORMATION		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	40	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	250	mBar	Temperature	60 °C
		Liquid Flow at BASE Temperature	34.8	m3/h	Inlet Pressure	3.5 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	3.432 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	1.017	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.972	----	Compressibility Z	
ORIFICE PLATE		Bore Type: Concentric			Venturi Type:	
		Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:	
		Bore Outlet Style: Bevel			Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:	
		Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)	
		Gasket Contact Face Finish:			Pressure Tap Orientation:	
		Holding Ring Type:			Insertion Type:	
		Holding Ring Material:			Sensor Configuration:	
ORIFICE FLANGE		Orifice Plate Drawing: 6-111			Sensor Conn Size: Type: Class:	
		End Conn Size: 4" Type: Flanged Class: 300			Flange Facing: Flange Face Finish:	
		Flange Facing: Raised Face Flange Face Finish:			Sensor/Conn Material:	
		Pressure Tap Conn Size: 1/2" Type: NPT			Sensor/Conn Style:	
		Pressure Tap Style: Flange			Other Wetted Components Material:	
		Gasket Style:			Inst Conn Size: Inst Conn Mat:	
		Pressure Tap Orientation:			Inst Valve Type:	
		Meter Run Nominal Size:			Accuracy:	
PREFABRICATED METER RUN		End Conn Size: Type: Class:			Repeatability:	
		Orifice Flange Conn Size: Class:			Model Number:	
		Flange Facing: Flange Face Finish:			Manufacturer:	
		Pressure Tap Style:			Pipe Nominal Size: 4"	
		Pressure Tap Conn Size: Type:			Estimated Inside Pipe Diameter:	
		Pipe Schedule: Pipe Bore Diameter:			Straight Run Requirement: UOP Std. Dwg. 6-112	
		Upstream Length: Downstream Length:			Liquid-Base Temperature: 15°C	
		Pipe/Orifice Flange/End Conn Material:			Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Bolting/Nut Material:			Calculation Method:	
		Gasket Style: Gasket Material:			Primary Flow Element Calibration:	
REMARKS		Deleted.				
		Design Pressure : 8.86 kg/cm2(g) / F.V				
		Design Temperature : 250 degC				

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **12** Quantity: **1** Tag Number: **FE-23-1202**

Service: **Decomposer Feed Startup Bypass**

Sizing Information		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	80	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	250	mBar	Temperature	60 °C
		Liquid Flow at BASE Temperature	59.9	m3/h	Inlet Pressure	3.4 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	3.432 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	1.017	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.972	----	Compressibility Z	
ORIFICE PLATE		Bore Type: Eccentric §			Venturi Type:	
		Bore Inlet Edge Style: Square			Venturi Conn Size: Type: Class:	
		Bore Outlet Style:			Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:	
		Orifice Plate Material: 304 or 316 Stainless Steel			Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)	
		Gasket Contact Face Finish:			Pressure Tap Orientation:	
		Holding Ring Type:			Insertion Type:	
		Holding Ring Material:			Sensor Configuration:	
ORIFICE FLANGE		Orifice Plate Drawing: §			Sensor Conn Size: Type: Class:	
		End Conn Size: 6" Type: Flanged Class: 300			Flange Facing: Flange Face Finish:	
		Flange Facing: Raised Face Flange Face Finish:			Sensor/Conn Material:	
		Pressure Tap Conn Size: 1/2" Type: NPT			Sensor/Conn Style:	
		Pressure Tap Style: Flange			Other Wetted Components Material:	
		Gasket Style:			Inst Conn Size: Inst Conn Mat:	
		Pressure Tap Orientation:			Inst Valve Type:	
		Meter Run Nominal Size:			Accuracy:	
PREFABRICATED METER RUN		End Conn Size: Type: Class:			Repeatability:	
		Orifice Flange Conn Size: Class:			Model Number:	
		Flange Facing: Flange Face Finish:			Manufacturer:	
		Pressure Tap Style:			Pipe Nominal Size: 6"	
		Pressure Tap Conn Size: Type:			Estimated Inside Pipe Diameter:	
		Pipe Schedule: Pipe Bore Diameter:			Straight Run Requirement: §	
		Upstream Length: Downstream Length:			Liquid-Base Temperature: 15°C Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Pipe/Orifice Flange/End Conn Material:				
		Bolting/Nut Material:			Calculation Method: Primary Flow Element Calibration:	
		Gasket Style: Gasket Material:				
	Pressure Tap Orientation:					
	Manufacturer:					
REMARKS		§ If the orifice plate is located in a horizontal meter run, the orifice plate shall be fabricated in accordance with UOP Standard Drawing 6-111, Detail 'A', except that the orifice plate shall be the eccentric type. If reducers are used in the horizontal meter run they shall be eccentric and installed flat side down, and the upstream straight meter run shall be in accordance with UOP Standard Drawing 6-112.				
		Design Pressure : 10.0 kg/cm2(g) / F.V				
		Design Temperature : 250 degC				

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PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number **13** Quantity: **1** Tag Number: **FE-23-1205**
Service: **Flash Column Bottoms to Decomposer Drum**

Sizing Information		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	80	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	125	mBar	Temperature	60 °C
		Liquid Flow at BASE Temperature	59.9	m3/h	Inlet Pressure	3.4 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	3.432 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	1.017	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.972	----	Compressibility Z	
ORIFICE PLATE	Bore Type: Eccentric §				Venturi Type:	
	Bore Inlet Edge Style: Square				Venturi Conn Size: Type: Class:	
	Bore Outlet Style:				Flange Facing: Flange Face Finish:	
	Orifice Bore Diameter: Beta Ratio (d/D):				Venturi Material:	
	Orifice Plate Material: 304 or 316 Stainless Steel				Venturi Wall Thickness:	
	Plate Thickness: Plate Outside Diameter:				Pressure Tap Conn Size: Type:	
	Vent/Drain Hole Size: Vent/Drain Location:				Beta Ratio (d/D)	
	Gasket Contact Face Finish:				Pressure Tap Orientation:	
	Holding Ring Type:				Insertion Type:	
	Holding Ring Material:				Sensor Configuration:	
ORIFICE FLANGE	Orifice Plate Drawing: §				Sensor Conn Size: Type: Class:	
	End Conn Size: 6" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:	
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:	
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:	
	Pressure Tap Style: Flange				Other Wetted Components Material:	
	Gasket Style:				Inst Conn Size: Inst Conn Mat:	
	Pressure Tap Orientation:				Inst Valve Type:	
	Meter Run Nominal Size:				Accuracy:	
	End Conn Size: Type: Class:				Repeatability:	
	Orifice Flange Conn Size: Class:				Model Number:	
PREFABRICATED METER RUN	Flange Facing: Flange Face Finish:				Manufacturer:	
	Pressure Tap Style:				Pipe Nominal Size: 6"	
	Pressure Tap Conn Size: Type:				Estimated Inside Pipe Diameter:	
	Pipe Schedule: Pipe Bore Diameter:				Straight Run Requirement: §	
	Upstream Length: Downstream Length:				Liquid-Base Temperature: 15°C	
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
	Bolting/Nut Material:				Calculation Method:	
	Gasket Style: Gasket Material:				Primary Flow Element Calibration:	
	Pressure Tap Orientation:					
	Manufacturer:					
REMARKS	§ If the orifice plate is located in a horizontal meter run, the orifice plate shall be fabricated in accordance with UOP Standard Drawing 6-111, Detail 'A', except that the orifice plate shall be the eccentric type. If reducers are used in the horizontal meter run they shall be eccentric and installed flat side down, and the upstream straight meter run shall be in accordance with UOP Standard Drawing 6-112.					
	Design Pressure : 8.66 kg/cm2(g) / F.V Design Temperature : 195 degC					

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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PROJECT SPECIFICATION

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REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

Item Number **14** Quantity: **1** Tag Number: **FE-23-1201**
 Service: **Process Water to Decomposer Feed**

SIZING INFORMATION		Process Data Case	Single Case	Unit	Sizing Case	Unit
SIZING INFORMATION		Meter Maximum at BASE Condition	0.8	m3/h	Vapor Specific Gravity (Air=1)	----
		Differential Press Inst Calibration	250	mBar	Temperature	50 °C
		Liquid Flow at BASE Temperature	0.38	m3/h	Inlet Pressure	5.4 kg/cm2(g)
		Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	0.547 cP
		Steam Flow		kg/h	Vapor Viscosity	cP
		Liq Specific Gravity at BASE Temp	1.000	----	Specific Heats Cp/Cv (Real)	
		Liq Specific Gravity at Flow Temp	0.989	----	Compressibility Z	
ORIFICE PLATE		Bore Type:			Venturi Type:	
		Bore Inlet Edge Style:			Venturi Conn Size: Type: Class:	
		Bore Outlet Style:			Flange Facing: Flange Face Finish:	
		Orifice Bore Diameter: Beta Ratio (d/D):			Venturi Material:	
		Orifice Plate Material:			Venturi Wall Thickness:	
		Plate Thickness: Plate Outside Diameter:			Pressure Tap Conn Size: Type:	
		Vent/Drain Hole Size: Vent/Drain Location:			Beta Ratio (d/D)	
		Gasket Contact Face Finish:			Pressure Tap Orientation:	
		Holding Ring Type:			Insertion Type:	
		Holding Ring Material:			Sensor Configuration:	
ORIFICE FLANGE		Orifice Plate Drawing:			Sensor Conn Size: Type: Class:	
		End Conn Size: Type: Class:			Flange Facing: Flange Face Finish:	
		Flange Facing: Flange Face Finish:			Sensor/Conn Material:	
		Pressure Tap Conn Size: Type:			Sensor/Conn Style:	
		Pressure Tap Style:			Other Wetted Components Material:	
		Gasket Style:			Inst Conn Size: Inst Conn Mat:	
		Pressure Tap Orientation:			Inst Valve Type:	
		Meter Run Nominal Size:			Accuracy:	
					Repeatability:	
					Model Number:	
PREFABRICATED METER RUN		End Conn Size: 1" Type: Flanged Class: 150			Manufacturer:	
		Orifice Flange Conn Size: \$ Class: \$			Pipe Nominal Size: 1"	
		Flange Facing: Raised Face Flange Face Finish:			Estimated Inside Pipe Diameter:	
		Pressure Tap Style:			Straight Run Requirement:	
		Pressure Tap Conn Size: Type:			Liquid-Base Temperature: 15°C	
		Pipe Schedule: \$ Pipe Bore Diameter: \$			Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	
		Upstream Length: \$ Downstream Length: \$			Calculation Method:	
		Pipe/Orifice Flange/End Conn Material: 316 Stainless Steel			Primary Flow Element Calibration:	
		Bolting/Nut Material:				
		Gasket Style: \$ Gasket Material: \$				
REMARKS		§ Provide a 1 inch honed flow section complete with orifice flanges, 316 stainless steel concentric orifice plate, orifice flange gaskets, and end connections. Orifice flange gaskets shall be spiral wound per API 601 (except for vendor required dimensional tolerances) with stainless steel winding and flexible graphite filler.				
		Design Pressure : 9.716 kg/cm2(g)				
		Design Temperature : 120 degC				

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PROJECT SPECIFICATION

963766 — 605

PRIMARY FLOW ELEMENTS
(DIFFERENTIAL PRESSURE TYPE)

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	18SEP12	RED	MW				

Item Number **15** Quantity: **1** Tag Number: **FE-23-1301**Service: **Decomposer Product to Dehydrators**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit
		Meter Maximum at BASE Condition	85	m3/h		Vapor Specific Gravity (Air=1)	----
	Differential Press Inst Calibration	250	mBar		Temperature	83 °C	
	Liquid Flow at BASE Temperature	64.6	m3/h	1	Inlet Pressure	9.3 kg/cm2(g)	
	Vapor Flow at Base Condition		normal m3/h		Liquid Viscosity	0.532 cP	
	Steam Flow		kg/h		Vapor Viscosity	cP	
	Liq Specific Gravity at BASE Temp	0.949	----		Specific Heats Cp/Cv (Real)		
	Liq Specific Gravity at Flow Temp	0.892	----		Compressibility Z		

ORIFICE PLATE	ORIFICE FLANGE		PREFABRICATED METER RUN	VENTURI	INSERTION	INSTAL-LATION	BASE	SPECIAL
	Bore Type: Concentric Bore Inlet Edge Style: Square Bore Outlet Style: Bevel Orifice Bore Diameter: Beta Ratio (d/D): Orifice Plate Material: 304 or 316 Stainless Steel Plate Thickness: Plate Outside Diameter: Vent/Drain Hole Size: Vent/Drain Location: Gasket Contact Face Finish: Holding Ring Type: Holding Ring Material: Orifice Plate Drawing: 6-111	End Conn Size: 6" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Pressure Tap Conn Size: 1/2" Type: NPT Pressure Tap Style: Flange Gasket Style: Pressure Tap Orientation: Meter Run Nominal Size:						

REMARKS

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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 PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)				PROJECT SPECIFICATION 963766 — 605 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																				
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																
0	25JUN12	MVD	MW																																				
Item Number 16 Quantity: 1 Tag Number: FE-23-1401 Service: S4 Steam to Dehydrators																																							
SIZING INFORMATION	Process Data Case		Design	Unit	Sizing Case		Unit																																
	Meter Maximum at BASE Condition		4500	kg/h	Vapor Specific Gravity (Air=1)		0.622																																
	Differential Press Inst Calibration		125	mBar	Temperature		160																																
	Liquid Flow at BASE Temperature			m3/h	Inlet Pressure		4.0																																
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity																																		
	Steam Flow		3421	kg/h	Vapor Viscosity		0.015																																
	Liq Specific Gravity at BASE Temp			----	Specific Heats Cp/Cv (Real)		1.298																																
	Liq Specific Gravity at Flow Temp			----	Compressibility Z		0.960																																
ORIFICE PLATE	Bore Type: Concentric			VENTURI	Venturi Type:																																		
	Bore Inlet Edge Style: Square				Venturi Conn Size: Type: Class:																																		
	Bore Outlet Style: Bevel				Flange Facing: Flange Face Finish:																																		
	Orifice Bore Diameter: Beta Ratio (d/D):				Venturi Material:																																		
	Orifice Plate Material: 304 or 316 Stainless Steel				Venturi Wall Thickness:																																		
	Plate Thickness: Plate Outside Diameter:				Pressure Tap Conn Size: Type:																																		
	Vent/Drain Hole Size: Vent/Drain Location:				Beta Ratio (d/D)																																		
	Gasket Contact Face Finish:				Pressure Tap Orientation:																																		
	Holding Ring Type:				Insertion Type:																																		
	Holding Ring Material:				Sensor Configuration:																																		
ORIFICE FLANGE	Orifice Plate Drawing: 6-111			INSERTION	Sensor Conn Size: Type: Class:																																		
	End Conn Size: 8" Type: Flanged Class: 300				Flange Facing: Flange Face Finish:																																		
	Flange Facing: Raised Face Flange Face Finish:				Sensor/Conn Material:																																		
	Pressure Tap Conn Size: 1/2" Type: NPT				Sensor/Conn Style:																																		
	Pressure Tap Style: Flange				Other Wetted Components Material:																																		
	Gasket Style:				Inst Conn Size: Inst Conn Mat:																																		
	Pressure Tap Orientation:				Inst Valve Type:																																		
	Meter Run Nominal Size:				Accuracy:																																		
PREFABRICATED METER RUN	End Conn Size: Type: Class:			INSTAL-LATION	Repeatability:																																		
	Orifice Flange Conn Size: Class:				Model Number:																																		
	Flange Facing: Flange Face Finish:				Manufacturer:																																		
	Pressure Tap Style:				Pipe Nominal Size: 8"																																		
	Pressure Tap Conn Size: Type:			Estimated Inside Pipe Diameter:																																			
	Pipe Schedule: Pipe Bore Diameter:			Straight Run Requirement: UOP Std. Dwg. 6-112																																			
	Upstream Length: Downstream Length:			SPECIAL	Liquid-Base Temperature: 15°C																																		
	Pipe/Orifice Flange/End Conn Material:				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C																																		
	Bolting/Nut Material:				Calculation Method:																																		
	Gasket Style: Gasket Material:			Primary Flow Element Calibration:																																			
Pressure Tap Orientation:																																							
Manufacturer:																																							
REMARKS	Design Pressure : 16.5 kg/cm2(g) / F.V																																						
	Design Temperature : 195 degC																																						
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																							

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 PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)				PROJECT SPECIFICATION 963766 — 605 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td>1</td><td>28AUG12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>								REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW					1	28AUG12	MVD	MW																				
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																												
0	25JUN12	MVD	MW																																																
1	28AUG12	MVD	MW																																																
Item Number 17 Quantity: 1 Tag Number: FE-23-1701 Service: Reliable Cooling Water to Decomposer Cooler																																																			
SIZING INFORMATION	Process Data Case		Design		Unit		Sizing Case			Unit																																									
	Meter Maximum at BASE Condition		4500		m3/h		Vapor Specific Gravity (Air=1)			----																																									
	Differential Press Inst Calibration				mBar		Temperature			37 °C																																									
	Liquid Flow at BASE Temperature		3370		m3/h		Inlet Pressure			4.9 kg/cm2(g)																																									
	Vapor Flow at Base Condition				normal m3/h		Liquid Viscosity			0.692 cP																																									
	Steam Flow				kg/h		Vapor Viscosity			cP																																									
	Liq Specific Gravity at BASE Temp		1.000		----		Specific Heats Cp/Cv (Real)																																												
	Liq Specific Gravity at Flow Temp		0.994		----		Compressibility Z																																												
ORIFICE PLATE	Bore Type:				VENTURI	Venturi Type:																																													
	Bore Inlet Edge Style:					Venturi Conn Size:		Type:		Class:																																									
	Bore Outlet Style:					Flange Facing:		Flange Face Finish:																																											
	Orifice Bore Diameter:		Beta Ratio (d/D):			Venturi Material:																																													
	Orifice Plate Material:					Venturi Wall Thickness:																																													
	Plate Thickness:		Plate Outside Diameter:			Pressure Tap Conn Size:		Type:																																											
	Vent/Drain Hole Size:		Vent/Drain Location:			Beta Ratio (d/D)																																													
	Gasket Contact Face Finish:					Pressure Tap Orientation:																																													
	Holding Ring Type:					Insertion Type: ANNUBAR Diamond II (or equal) Sensor Configuration: 8 ports upstream and downstream Sensor Conn Size: S Type: Flange Class: 150 Flange Facing: Raised Face Flange Face Finish: Sensor/Conn Material: 316 Stainless Steel Sensor/Conn Style: Other Wetted Components Material: 316 Stainless Steel Inst Conn Size: 1/2" Inst Conn Mat: 316 SS Inst Valve Type: O, S & Y gate valves Accuracy: ±1% of flow rate Repeatability: ±0.1% Model Number: Manufacturer: Dieterich Standard (or equal)																																													
	Holding Ring Material:																																																		
ORIFICE FLANGE	Orifice Plate Drawing:				INSERTION																																														
	End Conn Size:		Type: Class:																																																
	Flange Facing:		Flange Face Finish:																																																
	Pressure Tap Conn Size:		Type:																																																
	Pressure Tap Style:																																																		
	Gasket Style:																																																		
	Pressure Tap Orientation:																																																		
	Meter Run Nominal Size:																																																		
PREFABRICATED METER RUN	End Conn Size:		Type: Class:		INSTALLATION	1 Pipe Nominal Size: 24"																																													
	Orifice Flange Conn Size:		Class:			Estimated Inside Pipe Diameter:																																													
	Flange Facing:		Flange Face Finish:			Straight Run Requirement:																																													
	Pressure Tap Style:																																																		
	Pressure Tap Conn Size:		Type:																																																
	Pipe Schedule:		Pipe Bore Diameter:																																																
	Upstream Length:		Downstream Length:																																																
	Pipe/Orifice Flange/End Conn Material:																																																		
	Bolting/Nut Material:																																																		
	Gasket Style:		Gasket Material:																																																
REMARKS	Pressure Tap Orientation:				SPECIAL	Liquid-Base Temperature: 15°C Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C																																													
	Manufacturer:					Calculation Method:																																													
						Primary Flow Element Calibration:																																													
§ Vendor to determine Sensor Connection Size Vendor to determine Differential Pressure Instrument Calibration required. Design Pressure : 13.0 kg/cm2(g) Design Temperature : 120 degC																																																			
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																			



PROJECT SPECIFICATION

963766 — 605

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)

Item Number **18** Quantity: **2** Tag Number: **FE-23-1801, 1802**
 Service: **Decomposer Circulation Liquid to Calorimeter**

SIZING INFORMATION	Process Data Case		Single Case	Unit	Sizing Case		Unit	
	Meter Maximum at BASE Condition		0.18	m3/h	Vapor Specific Gravity (Air=1)		----	
	Differential Press Inst Calibration		250	mBar	Temperature		79 °C	
	Liquid Flow at BASE Temperature		0.144	m3/h	Inlet Pressure		3.9 kg/cm2(g)	
	Vapor Flow at Base Condition			normal m3/h	Liquid Viscosity		0.571 cP	
	Steam Flow			kg/h	Vapor Viscosity		cP	
	Liq Specific Gravity at BASE Temp		0.949	----	Specific Heats Cp/Cv (Real)			
	Liq Specific Gravity at Flow Temp		0.898	----	Compressibility Z			
ORIFICE PLATE	Bore Type:			VENTURI	Venturi Type:			
	Bore Inlet Edge Style:				Venturi Conn Size: Type: Class:			
	Bore Outlet Style:				Flange Facing: Flange Face Finish:			
	Orifice Bore Diameter: Beta Ratio (d/D):				Venturi Material:			
	Orifice Plate Material:				Venturi Wall Thickness:			
	Plate Thickness: Plate Outside Diameter:				Pressure Tap Conn Size: Type:			
	Vent/Drain Hole Size: Vent/Drain Location:				Beta Ratio (d/D)			
	Gasket Contact Face Finish:				Pressure Tap Orientation:			
	Holding Ring Type:				INSERTION			
	Holding Ring Material:							
Orifice Plate Drawing:			Insertion Type:					
ORIFICE FLANGE	End Conn Size: Type: Class:			INSERTION	Sensor Configuration:			
	Flange Facing: Flange Face Finish:				Sensor Conn Size: Type: Class:			
	Pressure Tap Conn Size: Type:				Flange Facing: Flange Face Finish:			
	Pressure Tap Style:				Sensor/Conn Material:			
	Gasket Style:				Sensor/Conn Style:			
	Pressure Tap Orientation:				Other Wetted Components Material:			
	Meter Run Nominal Size:				Inst Conn Size: Inst Conn Mat:			
					Inst Valve Type:			
PREFABRICATED METER RUN	1 End Conn Size: 3/4" Type: Flanged Class: 300			INSTAL- LATION	Accuracy:			
	Orifice Flange Conn Size: \$ Class: \$				Repeatability:			
	Flange Facing: Raised Face Flange Face Finish:				Model Number:			
	Pressure Tap Style:				Manufacturer:			
	Pressure Tap Conn Size: Type:			BASE	Pipe Nominal Size: 3/4"			
	Pipe Schedule: \$ Pipe Bore Diameter: \$				Estimated Inside Pipe Diameter:			
	Upstream Length: \$ Downstream Length: \$				Straight Run Requirement:			
	1 Pipe/Orifice Flange/End Conn Material: 316L Stainless Steel							
	REMARKS	Bolting/Nut Material:			SPECIAL	Liquid-Base Temperature: 15°C		
		Gasket Style: \$ Gasket Material: \$				Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C		
Pressure Tap Orientation:			Calculation Method:					
Manufacturer: Daniel (or equal)			Primary Flow Element Calibration:					
§ Provide a 3/4 inch honed flow section complete with orifice flanges, 316 stainless steel concentric orifice plate, orifice flange gaskets, and end connections. Orifice flange gaskets shall be spiral wound per API 601 (except for vendor required dimensional tolerances) with stainless steel winding and flexible graphite filler.								
1 Provide a 316L Stainless Steel 'Y' type strainer with 3/4 inch Class 300 RF flanges and a stainless steel mesh with mesh size suitable for the calculated orifice bore.								
1								
Design Pressure : 16.5 kg/cm2(g) / F.V								
Design Temperature : 250 degC								

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					PROJECT SPECIFICATION 963766 — 605							
PRIMARY FLOW ELEMENTS (DIFFERENTIAL PRESSURE TYPE)					REV	DATE	BY	APVD	REV	DATE	BY	APVD
					0	25JUN12	MVD	MW				

Item	Item Number 19 Quantity: 1	Tag Number: FE-23-2001
	Service: Acid Aromatics to Direct Neutralizer Static Mixer	

SIZING INFORMATION	Process Data Case	Single Case	Unit	Sizing Case	Unit
	Meter Maximum at BASE Condition	30	m3/h	Vapor Specific Gravity (Air=1)	----
	Differential Press Inst Calibration	250	mBar	Temperature	38
	Liquid Flow at BASE Temperature	21.6	m3/h	Inlet Pressure	6.3
	Vapor Flow at Base Condition		normal m3/h	Liquid Viscosity	1.160
	Steam Flow		kg/h	Vapor Viscosity	cP
	Liq Specific Gravity at BASE Temp	0.949	----	Specific Heats Cp/Cv (Real)	
	Liq Specific Gravity at Flow Temp	0.939	----	Compressibility Z	

ORIFICE PLATE	ORIFICE FLANGE	VENTURI	INSERTION
Bore Type: Concentric Bore Inlet Edge Style: Square Bore Outlet Style: Bevel Orifice Bore Diameter: Beta Ratio (d/D): Orifice Plate Material: 304 or 316 Stainless Steel Plate Thickness: Plate Outside Diameter: Vent/Drain Hole Size: Vent/Drain Location: Gasket Contact Face Finish: Holding Ring Type: Holding Ring Material: Orifice Plate Drawing: 6-111	End Conn Size: 3" Type: Flanged Class: 300 Flange Facing: Raised Face Flange Face Finish: Pressure Tap Conn Size: 1/2" Type: NPT Pressure Tap Style: Flange Gasket Style: Pressure Tap Orientation: Meter Run Nominal Size:	Venturi Type: Venturi Conn Size: Type: Class: Flange Facing: Flange Face Finish: Venturi Material: Venturi Wall Thickness: Pressure Tap Conn Size: Type: Beta Ratio (d/D) Pressure Tap Orientation:	Insertion Type: Sensor Configuration: Sensor Conn Size: Type: Class: Flange Facing: Flange Face Finish: Sensor/Conn Material: Sensor/Conn Style: Other Wetted Components Material: Inst Conn Size: Inst Conn Mat: Inst Valve Type: Accuracy: Repeatability: Model Number: Manufacturer:
PREFABRICATED METER RUN	INSTAL-LATION	BASE	SPECIAL
End Conn Size: Type: Class: Orifice Flange Conn Size: Class: Flange Facing: Flange Face Finish: Pressure Tap Style: Pressure Tap Conn Size: Type: Pipe Schedule: Pipe Bore Diameter: Upstream Length: Downstream Length: Pipe/Orifice Flange/End Conn Material: Bolting/Nut Material: Gasket Style: Gasket Material: Pressure Tap Orientation: Manufacturer:	Pipe Nominal Size: 3" Estimated Inside Pipe Diameter: Straight Run Requirement: UOP Std. Dwg. 6-112	Liquid-Base Temperature: 15°C Vapor-Base Press: 1.033 kg/cm2(a) , Base Temp: 0°C	Calculation Method: Primary Flow Element Calibration:

REMARKS	Design Pressure : 21.0 kg/cm2(g) Design Temperature : 250 degC

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

Revision
Indication

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MASS FLOW METERS

PROJECT SPECIFICATION

963766 — 642

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Item	Quantity	Description
1	1	Provide a vibrating tube, Coriolis type mass flow meter complete with sensor, transmitter, and local indicator. Transmitter output shall be 4 to 20 mA with 20 mA corresponding to the meter maximum and 4 mA corresponding to zero flow. Ambient temperature range is 15 to 40°C. Tag Number: FT-23-1003 Service: Sealant Return from Concentration Vacuum Producing Equipment Sensor Data Process connections: 1/2" 150 RF Process Wetted Parts: Sensor Tube – 316L Stainless steel Flanges – 316L Stainless steel Accuracy: at least $\pm 0.3\%$ of rate at operating temperature Operating Data Fluid: Water Flowing fluid is not abrasive. Normal Flow: 1791 kg/h Meter Maximum: 2300 kg/h Sp. Gr. @ Flow Temp: 0.997 Temperature: 18°C , Design Temperature : 250 degC Pressure: 1.9 kg/cm2(g) , Design Pressure : 4.0 kg/cm2(g), F.V Viscosity @ Flow Temp: 1.073 Allowable ΔP: 0.49 kg/cm2 Installation Installation shall be in accordance with vendor's recommendations. Process piping to the flow sensor shall be properly aligned and supported to avoid excessive stress on the flow sensor. Utilities Available Electrical Power: 220 Volts, 50 Hz. Supplier (or equal): Micro Motion, Inc. 7070 Winchester Circle Boulder, Colorado 80301 USA www.emersonprocess.com Endress + Hauser, Inc. 2350 Endress Place PO Box 246 Greenwood, Indiana 46143 USA www.endress.com

Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification: 601, this project

Revision
Indication

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MASS FLOW METERS

PROJECT SPECIFICATION

963766 — 642

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Item	Quantity	Description
2	1	Provide a vibrating tube, Coriolis type mass flow meter complete with sensor, transmitter, and local indicator. Transmitter output shall be 4 to 20 mA with 20 mA corresponding to the meter maximum and 4 mA corresponding to zero flow. Ambient temperature range is 15 to 40°C. Tag Number: FT-23-1002 Service: Process Water to Concentration Vacuum Producing Equipment Sensor Data Process connections: 1/2" 150 RF Process Wetted Parts: Sensor Tube – 316L Stainless steel Flanges – 316L Stainless steel Accuracy: at least $\pm 0.3\%$ of rate at operating temperature Operating Data Fluid: Water Flowing fluid is not abrasive. Normal Flow: 227 kg/h Meter Maximum: 300 kg/h Sp. Gr. @ Flow Temp: 0.969 Temperature: 50°C , Design Temperature : 120 degC Pressure: 5.0 kg/cm2(g) , Design Pressure : 11.26 kg/cm2(g) Viscosity @ Flow Temp: 0.560 Allowable ΔP: 0.49 kg/cm2 Installation Installation shall be in accordance with vendor's recommendations. Process piping to the flow sensor shall be properly aligned and supported to avoid excessive stress on the flow sensor. Utilities Available Electrical Power: 220 Volts, 50 Hz. Supplier (or equal): Micro Motion, Inc. 7070 Winchester Circle Boulder, Colorado 80301 USA www.emersonprocess.com Endress + Hauser, Inc. 2350 Endress Place PO Box 246 Greenwood, Indiana 46143 USA www.endress.com

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Indication

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MASS FLOW METERS

PROJECT SPECIFICATION

963766 — 642

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description
3	3	Provide a vibrating tube, Coriolis type mass flow meter complete with sensor, transmitter, and local indicator. Transmitter output shall be 4 to 20 mA with 20 mA corresponding to the meter maximum and 4 mA corresponding to zero flow. Ambient temperature range is 15 to 40°C.

Tag Number: FXT-23-1501, FXT-23-1601, FXT-23-1602

Service: Decomposer Acid Injection

Sensor Data

Process connections:	1/4" Compression fittings
Process Wetted Parts:	Sensor Tube – Manufacturer's Recommendation Connections – Manufacturer's Recommendation
Accuracy:	at least $\pm 0.3\%$ of rate at operating temperature

Operating Data

Fluid:	Concentrated Sulfuric Acid Flowing fluid is not abrasive.
Normal Flow:	0.82 kg/h
Meter Maximum:	2.0 kg/h
Sp. Gr. @ Flow Temp:	1.715
Temperature:	30°C , Design Temperature : 250 degC
Pressure:	2.7 kg/cm2(g) , Design Pressure : 4.921 kg/cm2(g)
Viscosity @ Flow Temp:	14.174
Allowable ΔP :	0.49 kg/cm2

Installation

Installation shall be in accordance with vendor's recommendations. Process piping to the flow sensor shall be properly aligned and supported to avoid excessive stress on the flow sensor.

Utilities Available

Electrical Power: 220 Volts, 50 Hz.

Supplier (or equal):

Micro Motion, Inc.
7070 Winchester Circle
Boulder, Colorado 80301 USA
www.emersonprocess.com

Endress + Hauser, Inc.
2350 Endress Place
PO Box 246
Greenwood, Indiana 46143 USA
www.endress.com

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MASS FLOW METERS

PROJECT SPECIFICATION

963766 — 642

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description
4	1	Provide a vibrating tube, Coriolis type mass flow meter complete with sensor, transmitter, and local indicator. Transmitter output shall be 4 to 20 mA with 20 mA corresponding to the meter maximum and 4 mA corresponding to zero flow. Ambient temperature range is 15 to 40°C. Tag Number: FT-23-1903 Service: Diamine Injection to Direct Neutralization Static Mixer Sensor Data Process connections: 1/2" 150 RF Process Wetted Parts: Sensor Tube – 304 / 316 stainless steel Flanges – 304 / 316 stainless steel Accuracy: at least $\pm 0.3\%$ of rate at operating temperature Operating Data Fluid: Diamine – Dytek (A) Flowing fluid is not abrasive. Normal Flow: 3 kg/h Meter Maximum: 6 kg/h Sp. Gr. @ Flow Temp: 0.856 Temperature: 31°C , Design Temperature : 120 degC Pressure: 4.8 kg/cm2(g) , Design Pressure : 6.546 kg/cm2(g) Viscosity @ Flow Temp: 2.233 Allowable ΔP: 0.49 kg/cm2 Installation Installation shall be in accordance with vendor's recommendations. Process piping to the flow sensor shall be properly aligned and supported to avoid excessive stress on the flow sensor. Utilities Available Electrical Power: 220 Volts, 50 Hz.

Supplier (or equal):

Micro Motion, Inc.
7070 Winchester Circle
Boulder, Colorado 80301 USA
www.emersonprocess.com

Endress + Hauser, Inc.
2350 Endress Place
PO Box 246
Greenwood, Indiana 46143 USA
www.endress.com

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A Honeywell Company


THERMAL MASS FLOW INSTRUMENTS

PROJECT SPECIFICATION

963766 — 689

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description
1	1	Single point in-line thermal convective mass flow meter. The flow measurement shall be in engineering units for gas flow referenced to standard conditions of 0°C and 1.033 kg/cm2(a). Tag Number: FT-23-1001 Service: Nitrogen to Concentration Vacuum Producing Equipment Operating Conditions Fluid State: Gas Fluid: Nitrogen Pressure: 7.0 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g) Temperature: 38°C , Design Temperature : 120 degC Specific Gravity: 0.967 Normal flow: 29 Nm3/h Mass Flow Element: The flow element shall be composed of a thermal mass flow sensor inserted into a pipe section (body) that has terminal connections for process line installation. Sensor Material: Alloy C276 Sensor Accuracy: ± 1% of reading Sensor Repeatability: 0.25% Sensor Response: 3 seconds Body Material: 316L stainless steel Body Size: 1 inches by 12 inches Body Connections: 1 inch Class 150 RF Flow Range: 0 - 40 Nm3/h Flow Transmitter: The flow transmitter shall be microprocessor based with front cover keypad and display, and shall incorporate sensor temperature limiting circuitry for combustible gases. The transmitter output shall be 4 - 20 mA. Available electrical power is 220 VAC, 50 Hz. Supplier (or equal) Kurz Instruments Inc. 2411 Garden Road Monterey, California 93940 USA www.kurz-instruments.com

 Other specifications
referenced in this
specification:

 UOP Standard Specification:
 UOP Standard Drawing:
 UOP Project Specification: 601, this project

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Indication

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THERMAL MASS FLOW INSTRUMENTS



PROJECT SPECIFICATION

963766 — 689

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Item	Quantity	Description																						
3	1	Insertion thermal dispersion type flow switch/monitor with built-in calibration circuit. Switch relay shall be hermetically sealed SPDT type. Contact rating by Contractor. Available electrical power is 220 VAC, 50 Hz. The switch relay shall deenergize and the contact shall open when flow falls below the setpoint. The switch response time and switch point shall be adjustable. The switch contact shall be wired to the logic system (See UOP Project Specification 660) Tag Number: FXSL-23-1203 Service: Cumene Flush to Oxidate Long Circulation Line Operating Conditions <table><tr><td>Fluid State:</td><td>Liquid</td></tr><tr><td>Fluid:</td><td>Cumene</td></tr><tr><td>Pressure:</td><td>1.6 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)</td></tr><tr><td>Temperature:</td><td>46°C , Design Temperature : 120 degC</td></tr><tr><td>Sp. Gr. @ Flow Temperature:</td><td>0.841</td></tr><tr><td>Normal flow:</td><td>13150 kg/h</td></tr></table> Mass Flow Element: The flow element shall be composed of a thermal mass flow sensor for insertion into the process line. <table><tr><td>Sensor Wetted Parts:</td><td>316L stainless steel</td></tr><tr><td>Sensor End Connection:</td><td>1 inch 150 RF, 316L SS flange assembly</td></tr><tr><td>Sensor Repeatability:</td><td>1 % of full scale</td></tr><tr><td>Sensor Response:</td><td>5 seconds (initial setting)</td></tr><tr><td>Switch point:</td><td>2630 kg/h (initial setting)</td></tr></table> Installation: Pipe Nominal Size: 3 inch Orientation: Installation in a vertical line with flow upward is preferred; however, installation in a horizontal line with the probe horizontal is also acceptable. Insertion Length: The vendor shall determine the insertion length required based on a 6" nozzle projection. For insulated lines and vessels, a longer nozzle projection may be required, depending on the insulation thickness. The Contractor shall coordinate with the vendor and shall confirm the nozzle projection.	Fluid State:	Liquid	Fluid:	Cumene	Pressure:	1.6 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)	Temperature:	46°C , Design Temperature : 120 degC	Sp. Gr. @ Flow Temperature:	0.841	Normal flow:	13150 kg/h	Sensor Wetted Parts:	316L stainless steel	Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly	Sensor Repeatability:	1 % of full scale	Sensor Response:	5 seconds (initial setting)	Switch point:	2630 kg/h (initial setting)
Fluid State:	Liquid																							
Fluid:	Cumene																							
Pressure:	1.6 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)																							
Temperature:	46°C , Design Temperature : 120 degC																							
Sp. Gr. @ Flow Temperature:	0.841																							
Normal flow:	13150 kg/h																							
Sensor Wetted Parts:	316L stainless steel																							
Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly																							
Sensor Repeatability:	1 % of full scale																							
Sensor Response:	5 seconds (initial setting)																							
Switch point:	2630 kg/h (initial setting)																							

Supplier (or equal):

Fluid Components Intl
 1755 La Costa Meadows Drive
 San Marcos, California 92078 USA
 www.fluidcomponents.com

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THERMAL MASS FLOW INSTRUMENTS



PROJECT SPECIFICATION

963766 — 689

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Item	Quantity	Description																						
4	1	Insertion thermal dispersion type flow switch/monitor with built-in calibration circuit. Switch relay shall be hermetically sealed SPDT type. Contact rating by Contractor. Available electrical power is 220 VAC, 50 Hz. The switch relay shall deenergize and the contact shall open when flow falls below the setpoint. The switch response time and switch point shall be adjustable. The switch contact shall be wired to the the logic system (See UOP Project Specification 660) Tag Number: FXSL-23-1206 Service: Cumene Flush to Decomposer Feed Line Operating Conditions <table><tr><td>Fluid State:</td><td>Liquid</td></tr><tr><td>Fluid:</td><td>Cumene</td></tr><tr><td>Pressure:</td><td>1.5 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)</td></tr><tr><td>Temperature:</td><td>46°C , Design Temperature : 120 degC</td></tr><tr><td>Sp. Gr. @ Flow Temperature:</td><td>0.841</td></tr><tr><td>Normal flow:</td><td>3150 kg/h</td></tr></table> Mass Flow Element: The flow element shall be composed of a thermal mass flow sensor for insertion into the process line. <table><tr><td>Sensor Wetted Parts:</td><td>316L stainless steel</td></tr><tr><td>Sensor End Connection:</td><td>1 inch 150 RF, 316L SS flange assembly</td></tr><tr><td>Sensor Repeatability:</td><td>1 % of full scale</td></tr><tr><td>Sensor Response:</td><td>5 seconds (initial setting)</td></tr><tr><td>Switch point:</td><td>630 kg/h (initial setting)</td></tr></table> Installation: Pipe Nominal Size: 1-1/2 inch Orientation: Installation in a vertical line with flow upward is preferred; however, installation in a horizontal line with the probe horizontal is also acceptable. Insertion Length: The vendor shall determine the insertion length required based on a 6" nozzle projection. For insulated lines and vessels, a longer nozzle projection may be required, depending on the insulation thickness. The Contractor shall coordinate with the vendor and shall confirm the nozzle projection.	Fluid State:	Liquid	Fluid:	Cumene	Pressure:	1.5 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)	Temperature:	46°C , Design Temperature : 120 degC	Sp. Gr. @ Flow Temperature:	0.841	Normal flow:	3150 kg/h	Sensor Wetted Parts:	316L stainless steel	Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly	Sensor Repeatability:	1 % of full scale	Sensor Response:	5 seconds (initial setting)	Switch point:	630 kg/h (initial setting)
Fluid State:	Liquid																							
Fluid:	Cumene																							
Pressure:	1.5 kg/cm2(g) , Design Pressure : 10.0 kg/cm2(g)																							
Temperature:	46°C , Design Temperature : 120 degC																							
Sp. Gr. @ Flow Temperature:	0.841																							
Normal flow:	3150 kg/h																							
Sensor Wetted Parts:	316L stainless steel																							
Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly																							
Sensor Repeatability:	1 % of full scale																							
Sensor Response:	5 seconds (initial setting)																							
Switch point:	630 kg/h (initial setting)																							

Supplier (or equal):

Fluid Components Intl
 1755 La Costa Meadows Drive
 San Marcos, California 92078 USA
 www.fluidcomponents.com

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THERMAL MASS FLOW INSTRUMENTS

PROJECT SPECIFICATION

963766 — 689

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description																						
5	1	Insertion thermal dispersion type flow switch/monitor with built-in calibration circuit. Switch relay shall be hermetically sealed SPDT type. Contact rating by Contractor. Available electrical power is 220 VAC, 50 Hz. The switch relay shall deenergize and the contact shall open when flow falls below the setpoint. The switch response time and switch point shall be adjustable. The switch contact shall be wired to the control center to actuate an alarm. Tag Number: FSL-23-1207 Service: Process Water to Decomposer Feed Line Operating Conditions <table><tr><td>Fluid State:</td><td>Liquid</td></tr><tr><td>Fluid:</td><td>Water</td></tr><tr><td>Pressure:</td><td>1.5 kg/cm2(g) , Design Pressure : 9.716 kg/cm2(g)</td></tr><tr><td>Temperature:</td><td>50°C , Design Temperature : 120 degC</td></tr><tr><td>Sp. Gr. @ Flow Temperature:</td><td>0.989</td></tr><tr><td>Normal flow:</td><td>375 kg/h</td></tr></table> Mass Flow Element: The flow element shall be composed of a thermal mass flow sensor for insertion into the process line. <table><tr><td>Sensor Wetted Parts:</td><td>316L stainless steel</td></tr><tr><td>Sensor End Connection:</td><td>1 inch 150 RF, 316L SS flange assembly</td></tr><tr><td>Sensor Repeatability:</td><td>1 % of full scale</td></tr><tr><td>Sensor Response:</td><td>5 seconds (initial setting)</td></tr><tr><td>Switch point:</td><td>75 kg/h (initial setting)</td></tr></table> Installation: Pipe Nominal Size: 1 inch Orientation: Installation in a vertical line with flow upward is preferred; however, installation in a horizontal line with the probe horizontal is also acceptable. Insertion Length: The vendor shall determine the insertion length required based on a 6" nozzle projection. For insulated lines and vessels, a longer nozzle projection may be required, depending on the insulation thickness. The Contractor shall coordinate with the vendor and shall confirm the nozzle projection.	Fluid State:	Liquid	Fluid:	Water	Pressure:	1.5 kg/cm2(g) , Design Pressure : 9.716 kg/cm2(g)	Temperature:	50°C , Design Temperature : 120 degC	Sp. Gr. @ Flow Temperature:	0.989	Normal flow:	375 kg/h	Sensor Wetted Parts:	316L stainless steel	Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly	Sensor Repeatability:	1 % of full scale	Sensor Response:	5 seconds (initial setting)	Switch point:	75 kg/h (initial setting)
Fluid State:	Liquid																							
Fluid:	Water																							
Pressure:	1.5 kg/cm2(g) , Design Pressure : 9.716 kg/cm2(g)																							
Temperature:	50°C , Design Temperature : 120 degC																							
Sp. Gr. @ Flow Temperature:	0.989																							
Normal flow:	375 kg/h																							
Sensor Wetted Parts:	316L stainless steel																							
Sensor End Connection:	1 inch 150 RF, 316L SS flange assembly																							
Sensor Repeatability:	1 % of full scale																							
Sensor Response:	5 seconds (initial setting)																							
Switch point:	75 kg/h (initial setting)																							

Supplier (or equal):

Fluid Components Intl
 1755 La Costa Meadows Drive
 San Marcos, California 92078 USA
www.fluidcomponents.com

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THERMAL MASS FLOW INSTRUMENTS

PROJECT SPECIFICATION

963766 — 689

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item	Quantity	Description
2	1	Single point in-line thermal convective mass flow meter. The flow measurement shall be in engineering units for gas flow referenced to standard conditions of 0°C and 1.033 kg/cm2(a). Tag Number: FT-23-1004 Service: Concentration Vacuum Producing Equipment Vent Operating Conditions Fluid State: Gas Fluid: Nitrogen with trace hydrocarbon Pressure: 0.07 kg/cm2(g) , Design Pressure : 4.5 kg/cm2(g), F.V Temperature: 20°C , Design Temperature : 120 degC Specific Gravity: 0.981 Normal flow: 73 Nm3/h Mass Flow Element: The flow element shall be composed of a thermal mass flow sensor inserted into a pipe section (body) that has terminal connections for process line installation. Sensor Material: Alloy C276 Sensor Accuracy: ± 1% of reading Sensor Repeatability: 0.25% Sensor Response: 3 seconds Body Material: 316L stainless steel Body Size: 2 inches by 24 inches Body Connections: 2 inch Class 150 RF Flow Range: 0 - 100 Nm3/h Flow Transmitter: The flow transmitter shall be microprocessor based with front cover keypad and display, and shall incorporate sensor temperature limiting circuitry for combustible gases. The transmitter output shall be 4 - 20 mA. Available electrical power is 220 VAC, 50 Hz. Supplier (or equal) Kurz Instruments Inc. 2411 Garden Road Monterey, California 93940 USA www.kurz-instruments.com

						FLOW INSTRUMENT PROCESS SPECIFICATION					
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED						PROJECT No: 120117					
LOCATION: MAP TA PHUT, THAILAND						UNIT No: 2300					
REV	A1	F1	F2	Z1	DOCUMENT No. 14780-8120-PS-0031						
DATE	APR 18, 2014	JUN 25, 2014	OCT 22, 2014	MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee							
1	General	Item No.		RO-23-0501		RO-23-0601		RO-23-0701		RO-23-0801	
		P&ID No.		14780-8120-25-23-0005A		14780-8120-25-23-0006		14780-8120-25-23-0007A		14780-8120-25-23-0008B	
		Service		Preflash Column Steam heater Condensate Pump Min. Flow Line		Cumene to Concentration Cumene Quench Drum		Flash Column Vaporizer Condensate Pump Min. Flow Line		Preflash And Flash Columns Overhead Pumps Start Up	
3	Line Size	inch		1 1/2"		2"		1 1/2"		6"	
4	Fluid Properties	Type									
		State		LIQ		LIQ		LIQ		LIQ	
		Description		Condensate		Cumene		Condensate		Cumene	
		Corrosion/ Erosion due to									
5	Orifice Size	mm		3		6		2		11	
7	Normal Flow	Vap. (Nm3/hr)									
		Steam & Mass flow (kg/hr)									
		Liq. (m3/hr)		3.30		1.46 (Note 3)		1.75		66.00	
8	Max. Flow %										
9	Min. Flow %										
	LIQUID PROPERTIES										
10	S.G. @ 15°C		1		0.865		1		0.866		
11	S.G. @ flow conditions		0.947		0.847		0.931		0.847		
12	Viscosity @ flow conditions cP		0.248		0.624		0.204		0.622		
13	Vapor pressure kg/cm ² (a)		1.81		0.93		3.23		0.02		
14	Liquid Critical Press kg/cm ² (a)		225.55		32.72		224.9		32.74		
15	Liquid Electric Conductivity mS/m										
	VAPOUR PROPERTIES										
16	Mol.Wt.										
17	Viscosity @ flow conditions cP										
18	Vapour Cp/Cv @ flow conditions										
19	Vapour Compressibility (PV/RT)										
20	Vapour Density kg/m ³										
21	Operating temp °C	Minimum									
		Normal		135		38		135		39	
		Maximum									
22	Pressures kg/cm ² g	Normal Flow		2.701	1.57	5	2.46	3.172	2.31	7.575	-0.296
23	Upstream / Downstream	Maximum Flow									
33	High Alarm	m3/hr		N/A		N/A		N/A		N/A	
34	Low Alarm	m3/hr		N/A		N/A		N/A		N/A	
35	Design Pressure	kg/cm ² (G)		9.233		10.59		9.233		13.55/FV	
36	Design Temperature	°C		195		120		195		120	
37	Remarks										
38	Revision										
NOTES 1. Deleted. 2. Deleted. 3. Estimated flow rate is based on orifice hole size specified by UOP.											

						FLOW INSTRUMENT PROCESS SPECIFICATION					
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED						PROJECT No: 120117					
LOCATION: MAP TA PHUT, THAILAND						UNIT No: 2300					
REV	A1	F1	F2	Z1	DOCUMENT No. 14780-8120-PS-0031						
DATE	APR 18, 2014	JUN 25, 2014	OCT 22, 2014	MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee							
1	General	Item No.		RO-23-0901		RO-23-1001		RO-23-1201		RO-23-1202	
		P&ID No.		14780-8120-25-23-0009		14780-8120-25-23-0010		14780-8120-25-23-0012		14780-8120-25-23-0012	
		Service		Flash Column Bottoms Pump Spill Back		Nitrogen to Concentration Vacuum Producing Equipment		CHP Line Cumene Flush		CHP Line drain	
3	Line Size	inch		4"		1"		3"		1 1/2"	
4	Fluid Properties	Type									
		State		LIQ		VAP		LIQ		LIQ	
		Description		CHP		Nitrogen		Cumene		CHP	
		Corrosion/ Erosion due to									
5	Orifice Size	mm		10		2		25		12	
7	Normal Flow	Vap. (Nm3/hr)				3(NOTE 3)					
		Steam & Mass flow (kg/hr)									
		Liq. (m3/hr)		34.80				16.05 (Note 3)		5.83 (Note 3)	
8	Max. Flow %										
9	Min. Flow %										
LIQUID PROPERTIES											
10	S.G. @ 15°C		1.016				0.865		1.016		
11	S.G. @ flow conditions		0.971				0.847		0.971		
12	Viscosity @ flow conditions cP		3.432				0.624		3.432		
13	Vapor pressure kg/cm ² (a)		0.01				0.93		0.01		
14	Liquid Critical Press kg/cm ² (a)		33.89				32.72		33.89		
15	Liquid Electric Conductivity mS/m										
VAPOUR PROPERTIES											
16	Mol.Wt.				28.01						
17	Viscosity @ flow conditions cP				0.0199						
18	Vapour Cp/Cv @ flow conditions				1.401						
19	Vapour Compressibility (PV/RT)				1						
20	Vapour Density kg/m ³				1.99						
21	Operating temp °C	Minimum									
		Normal		60		38		38		38	
		Maximum									
22	Pressures kg/cm ² g	Normal Flow		2.054	-0.99	0.84	0.21	1.6	0.7	3.4	0.7
23	Upstream / Downstream	Maximum Flow									
33	High Alarm	m3/hr		N/A		N/A		N/A		N/A	
34	Low Alarm	m3/hr		N/A		N/A		N/A		N/A	
35	Design Pressure	kg/cm ² (G)		10/FV		10		10		13.55	
36	Design Temperature	°C		250		120		120		250	
37	Remarks										
38	Revision										
NOTES 1. Deleted. 2. Deleted. 3. Estimated flow rate is based on orifice hole size specified by UOP.											

						FLOW INSTRUMENT PROCESS SPECIFICATION					
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED						PROJECT No: 120117					
LOCATION: MAP TA PHUT, THAILAND						UNIT No: 2300					
REV	A1	F1	F2	Z1	DOCUMENT No. 14780-8120-PS-0031						
DATE	APR 18, 2014	JUN 25, 2014	OCT 22, 2014	MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee							
1	General	Item No.		RO-23-1203		RO-23-2001		RO-23-2002		RO-23-2201	
		P&ID No.		14780-8120-25-23-0012A		14780-8120-25-23-0020A		14780-8120-25-23-0020A		14780-8120-25-23-0022	
		Service		Process Water to Decomposer Drum		Acid Aromatics Sump Pumps Min. Flow		Nitrogen to Acid Aromatics Sump		Nitrogen to Acid Aromatics Sub header Purge	
3	Line Size	inch		1"		2"		1"		1"	
4	Fluid Properties	Type									
		State		LIQ		LIQ		VAP		VAP	
		Description		Process Water		Hydrocarbon		Nitrogen		Nitrogen	
		Corrosion/ Erosion due to									
5	Orifice Size	mm		7		Vendor		3		3	
7	Normal Flow	Vap. (Nm3/hr)						6(NOTE 3)		6.00	
		Steam & Mass flow (kg/hr)									
		Liq. (m3/hr)		0.8(NOTE 3)		3.30					
8	Max. Flow %										
9	Min. Flow %										
	LIQUID PROPERTIES										
10	S.G. @ 15°C		1.00								
11	S.G. @ flow conditions		0.99		0.94						
12	Viscosity @ flow conditions cP		0.55		1.16						
13	Vapor pressure kg/cm ² (a)		0.13		1.18						
14	Liquid Critical Press kg/cm ² (a)		224.90		59.64						
15	Liquid Electric Conductivity mS/m										
	VAPOUR PROPERTIES										
16	Mol.Wt.						28.01		28.01		
17	Viscosity @ flow conditions cP						0.0199		0.0199		
18	Vapour Cp/Cv @ flow conditions						1.401		1.401		
19	Vapour Compressibility (PV/RT)						1		1		
20	Vapour Density kg/m ³						1.99		1.99		
21	Operating temp °C	Minimum									
		Normal		50		38		38		38	
		Maximum									
22	Pressures kg/cm ² g	Normal Flow		5.4	3.5	7.55	0.01	0.84	0.01	7	0.5
23	Upstream / Downstream	Maximum Flow									
33	High Alarm	m3/hr		N/A		N/A		N/A		N/A	
34	Low Alarm	m3/hr		N/A		N/A		N/A		N/A	
35	Design Pressure	kg/cm ² (G)		10.5		13.06		10		10	
36	Design Temperature	°C		120		325		120		120	
37	Remarks										
38	Revision										
NOTES 1. Deleted. 2. Deleted. 3. Estimated flow rate is based on orifice hole size specified by UOP.											

					FLOW INSTRUMENT PROCESS SPECIFICATION				
CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	A1	F1	F2	Z1	DOCUMENT No. 14780-8120-PS-0031				
DATE	APR 18, 2014	JUN 25, 2014	OCT 22, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.		FIF-23-2201					
		P&ID No.		14780-8120-25-23-0022					
				Nitrogen to Aromatics Relief Sub Header					
		Service							
3	Line Size	inch		1"					
4	Fluid Properties	Type							
		State		VAP					
		Description		Nitrogen					
		Corrosion/ Erosion due to							
5	Orifice Size	mm							
7	Normal Flow	Vap. (Nm3/hr)		6.00					
		Steam & Mass flow (kg/hr)							
		Liq. (m3/hr)							
8	Max. Flow %								
9	Min. Flow %								
	LIQUID PROPERTIES								
10	S.G. @ 15°C								
11	S.G. @ flow conditions								
12	Viscosity @ flow conditions cP								
13	Vapor pressure kg/cm ² (a)								
14	Liquid Critical Press kg/cm ² (a)								
15	Liquid Electric Conductivity mS/m								
	VAPOUR PROPERTIES								
16	Mol.Wt.		28.01						
17	Viscosity @ flow conditions cP		0.0199						
18	Vapour Cp/Cv @ flow conditions		1.401						
19	Vapour Compressibility (PV/RT)		1						
20	Vapour Density kg/m ³		1.99						
21	Operating temp °C	Minimum							
		Normal		AMB					
		Maximum							
22	Pressures kg/cm ² g	Normal Flow	7	(NOTE 1)					
23	Upstream / Downstream	Maximum Flow							
33	High Alarm	m3/hr	N/A						
34	Low Alarm	m3/hr	N/A						
35	Design Pressure	kg/cm ² (G)	10.5						
36	Design Temperature	°C	120						
37	Remarks								
38	Revision								
NOTES 1. Allowable pressure drop is 0.14 kg/cm ² at normal flow for this instrument.									